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A Feasibility Analysis of A Ferromagnetic Cerebral Stent as a Magnetic Retention Device For Localised Cell Therapeutics

Sam Lukaitis

(Supervisor: Alistair Elfick)

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Personal Statement

This individual project was initiated in January of 2026 and was a self-proposed concept with the help of my supervisor, Alistair Elfick and is not a continuation of any previous work. The concept of a magnetically active cerebral stent to enhance cell therapeutics was developed during discussions with Alistair Elfick building on ideas from the work on MSC transfection with mms6 genome and broader magnetic-targeting literature.

The specific engineering problem of effective capture range to allow for paracrine signalling, COMSOL analysis and python analytical framework were all developed independently. All figures, numerical results and written analysis in this report are my own and build upon fundamental physics grounded in literature which is derived and cited throughout this document.

The projects objectives and trajectory evolved throughout the semester. The scope narrowed and became focused on assessing feasibility of the concept through capture ranges after extensive literature research disproved initial 'cell propulsion' ideas. The implementation of the analytical framework to support the COMSOL results was driven by computational power restrictions and time availability.

I declare that this dissertation is my original work except where stated.

Signed: **SAM LUKAITIS**, Date: 20/04/2026

Abstract

Cellular therapies for the treatment of neurological disorders are primarily limited by inefficient delivery and retention of therapeutic cells and drugs at target sites. This project is a feasibility analysis of SPION-labelled cell capture by a ferromagnetic cerebral stent through trajectory-integrated analysis under an external $1.5T$ field, representative of an MRI environment. Magnetic gradient fields were evaluated across four stent geometries using COMSOL finite-element analysis and an Akoun-Yonnet analytical framework, which were cross-validated to within 1% of each other over physiologically relevant capture zones. The trajectory integration demonstrated that the static Furlani-Ng force-balance criterion, $|F_{\text{mag}}| > |F_{\text{drag}}|$, underestimates effective capture by 13.9x at healthy middle cerebral artery flow (0.6m/s mean velocity). The predicted capture range is $51 - 256\mu m$ across parametric sweeps of velocity and SPION loadings. The decay of the magnetic gradient field from a strut junction follows a power law with exponent $n \approx -1.247$ to -1.34, insensitive to strut count, underlining that radial capture performance is governed by strut magnetisation and geometry. The framework developed provides a quantitative foundation for further analysis and design optimisation, required to advance this feasibility framework into a physically complete and experimentally validated model.

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I would like to thank Alistair Elfick, my supervisor, for his support and guidance throughout this project, in particular for explaining biological concepts and assessing my ideas. I would also like to thank the School of Engineering for providing access to the COMSOL Multiphysics software and the University of Edinburgh for the access to literature resources.

Notation

The following tables define all mathematical symbols, acronyms, and specialised terminology used in this report.

Latin Symbols

Symbol	Description	Units
A	Power-law gradient prefactor at $d = 1\,\mathrm{mm}$	$\mathrm{T\,m^{-1}\,mm^{-n}}$
A_{100}	Power-law gradient evaluated at $d = 100\,\mu\mathrm{m}$	$\mathrm{T\,m^{-1}}$
$\mathbf{B}$	Magnetic flux density vector	T
B_0	Uniform external (MRI) magnetic field magnitude	T
$\mathbf{B}_{\mathrm{stent}}$	Magnetic flux density from stent struts only	T
$\mathbf{B}_{\mathrm{tot}}$	Total magnetic flux density ($\mathbf{B}_{\mathrm{stent}} + \mathbf{B}_0$)	T
$F(U, V, W)$	Akoun-Yonnet closed-form corner function	$\mathrm{m^2}$
F_{drag}	Hydrodynamic (Stokes) drag force on a cell	N
F_{mag}	Magnetic force on a SPION-labelled cell	N
F_p	Force parameter, $\lvert\mathbf{B}_{\mathrm{tot}}\rvert \cdot \nabla \lvert\mathbf{B}_{\mathrm{tot}}\rvert$	$\mathrm{T^2\,m^{-1}}$
$G(d)$	Magnetic field gradient magnitude as a function of distance	$\mathrm{T\,m^{-1}}$
$\mathbf{H}$	Magnetic field strength vector	$\mathrm{A\,m^{-1}}$
$L(\xi)$	Langevin function, $\coth(\xi) - 1/\xi$	—
M	Magnetisation vector magnitude	$\mathrm{A\,m^{-1}}$
M_{eff}	Effective (COMSOL-calibrated) strut magnetisation	$\mathrm{A\,m^{-1}}$
M_s	Saturation magnetisation of bulk magnetite	$\mathrm{A\,m^{-1}}$
R_i	3×3 rotation matrix for strut i	—
Re	Particle Reynolds number	—
T_{viscous}	Viscous relaxation time of a cell	s
U, V, W	Local prism-frame coordinates in Akoun-Yonnet derivation	m
V_{SPION}	Volume of internalised SPIONs per cell	$\mathrm{m^3}$
$\bar{v}$	Cross-sectional mean blood velocity	$\mathrm{m\,s^{-1}}$
v_{blood}	Local blood velocity vector at a point in the vessel	$\mathrm{m\,s^{-1}}$
v_{cell}	Cell velocity vector	$\mathrm{m\,s^{-1}}$
$v_z(r)$	Axial Poiseuille velocity profile	$\mathrm{m\,s^{-1}}$
m_{cell}	Cell mass	kg
m_{SPION}	SPION iron-oxide mass per cell	kg
n	Power-law decay exponent	—

Greek Symbols

Symbol	Description	Units
α	Womersley number, $R_{\text{ves}}\sqrt{\omega\rho/\eta}$	—
δ	Finite-difference step size; oscillatory boundary layer thickness	m
η	Dynamic viscosity of blood	$\text{Pa}\,\text{s}$
θ_i	Azimuthal angle of strut i	rad
μ_0	Permeability of free space, $4\pi \times 10^{-7}$	$\text{H}\,\text{m}^{-1}$
μ_r	Relative magnetic permeability of stent material	—
ξ	Dimensionless Langevin field parameter, $\alpha B/\mu_0$	—
ρ	Blood density	$\text{kg}\,\text{m}^{-3}$
ρ_{SPION}	Magnetite SPION material density	$\text{kg}\,\text{m}^{-3}$
σ_m	Surface magnetic charge density, $M\hat{n}$	$\text{A}\,\text{m}^{-1}$
τ_w	Wall shear stress	Pa
ϕ_m	Scalar magnetic potential	A
χ_0	Linear (unsaturated) SPION volume susceptibility	—
χ_{eff}	Langevin-corrected effective SPION susceptibility	—
η_{cap}	Population capture efficiency (fraction of cells captured)	—
ω	Angular frequency of cardiac pulsatility	$\text{rad}\,\text{s}^{-1}$

Acronyms and Abbreviations

Acronym	Definition
ACA	Anterior Cerebral Artery
AMT	Adsorptive-Mediated Transcytosis
BBB	Blood–Brain Barrier
BDNF	Brain-Derived Neurotrophic Factor
BCI	Brain–Computer Interface
CAD	Computer-Aided Design
CMT	Carrier-Mediated Transport
CNS	Central Nervous System
COMSOL	Commercial finite-element multiphysics simulation software (COMSOL Multiphysics)
FEM	Finite-Element Method
FGF-2	Fibroblast Growth Factor-2
FUS	Focused Ultrasound
MCA	Middle Cerebral Artery
mfnc	Magnetic Fields, No Currents (COMSOL physics module)
MRI	Magnetic Resonance Imaging
MSC	Mesenchymal Stem Cell
ODE	Ordinary Differential Equation
PCA	Posterior Cerebral Artery
RK45	Runge–Kutta 4th/5th order adaptive integrator
RMT	Receptor-Mediated Transcytosis
RMS	Root Mean Square
SPION	Superparamagnetic Iron Oxide Nanoparticle
STEP	Standard for the Exchange of Product model data (CAD file format)
VEGF	Vascular Endothelial Growth Factor

Risk Register

The risk register identifies the key risks associated with the modelling assumptions, numerical implementations and clinical translation for the proposed ferromagnetic cerebral stent concept presented in this report.

Category	**Risk Description**	**Mitigation / Control Measure**
Modelling	Steady Poiseuille flow assumption introduces error in the drag force calculation relative to pulsatile cerebral flow.	Identified as a first-order approximation and is reported in the limitations section. The oscillatory boundary layer thickness ($\delta \approx R/\alpha \approx 0.75\,\text{mm}$) is quantified to bound the approximation error.
Modelling	Strut-aligned cell injection ($\theta = 0$) overestimates population capture efficiency compared to injection across the full azimuth.	Results are presented as an upper bound throughout Sections 4.2.5 and 5.2. The reduction is estimated and discussed qualitatively.
Modelling	The single-ring analytical model negates the periodicity of the braided stent geometry, producing a conservative lower bound on effective capture range.	COMSOL cross-validation of the full three-dimensional geometry confirms the single-ring model captures the dominant capture physics. Acknowledged in the limitations section.
Numerical	COMSOL mesh singularities at strut corners produce non-physical gradient spikes at $d < 5$–$25\,\mu\text{m}$, corrupting near-surface data.	Model-specific radial cut-off distances implemented. Mesh independence confirmed: halving element count shifted threshold-crossing distances by less than 10µm.
Numerical	Linear relative permeability $\mu_r = 2$ does not implement the full B–H curve of a real ferromagnetic material, overestimating gradient at low B_0 and underestimating at deep saturation.	Value represents the saturated operating point at $B_0 = 1.5\,\text{T}$. The calibrated effective magnetisation $M_{\text{eff}} = 2.20\,\text{MA/m}$ absorbs the error through direct matching to the COMSOL gradient profile. A full B–H curve implementation is identified as a future refinement.

Category	**Risk Description**	**Mitigation / Control Measure**
Numerical	The Akoun–Yonnet superposition assumes each strut's magnetisation is unaffected by its neighbours; mutual induction between adjacent struts is not represented.	The COMSOL solution includes mutual induction. The analytical model is calibrated against this solution, absorbing the inter-strut coupling into M_{eff}.
Clinical	Wall shear stress not accounted for in modelling cell detachment, negating therapeutic impact even if the cell is captured.	Wall shear stress calculated and reported ($\tau_w \approx 2.1$). Detachment is acknowledged as a key limitation. A cell-surface adhesion model is identified as a required component of any full therapeutic retention analysis.
Clinical	Ferromagnetic material required for magnetic performance may introduce biocompatibility issues due to corrosion or leaching in coating-based approaches.	Material selection for a cerebrovascular ferromagnetic stent is unresolved within this analysis and is identified as a prerequisite for experimental translation.

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1 Introduction and Project Motivation

Stroke, traumatic brain injury and neurodegenerative diseases represent a large unmet therapeutic need, with strokes alone accounting for one in ten global deaths [1]. Cell-based therapies have significant potential in the treatment of neurological disorders, which are understood to act predominantly through paracrine signalling, the mechanism through which cells secrete growth factors and cytokines to suppress inflammation and promote repair mechanisms [2]. However, efficient delivery of stem cells (MSCs), neural progenitor cells, and drugs to the central nervous system remains a major challenge due to the highly restrictive nature of the blood-brain barrier (BBB) [3]. Therapies that bypass the BBB by direct intracranial injection face poor engraftment, with most injected cells dying or migrating within days [4, 5]. Magnetic targeting has emerged as a means of increasing targeting efficacy and has been shown to enhance cell delivery across the BBB when combined with transient disruption using MRI-guided focused ultrasound [6]. However, externally applied magnetic fields are constrained by tissue depth at clinically relevant levels and the rapid decay of magnetic gradients required for cell capture. Magnetic targeting of SPION-loaded cells addresses retention, however, external magnets approximately decay as $1/r^2$ in field and $1/r^3$ in gradient [7].

This project investigates the utilisation of a ferromagnetic cerebrovascular stent, saturated by a clinical MRI-strength external magnetic field to generate local $|\mathbf{B}|$ gradients of hundreds to thousands of T/m — capable of capturing SPION-labelled therapeutic cells and drugs at a target site under physiological flow conditions. The concept is demonstrated in peripheral and coronary arteries [8] with reported in-vivo capture efficiencies of 20%. This project is an analysis of cerebral feasibility, bridging the gap between minimally invasive devices and targeted neurotherapeutic delivery.

1.1 Objectives

The aim of this project is to define and analyse the capture range of a ferromagnetic stent in a cerebral artery and assess the feasibility of capturing magnetically labelled cells to enhance therapeutics in the brain.

Specifically:

- The review of relevant literature covering the BBB, paracrine signalling, magnetic labelling of cells and magnetically functional vascular implants, identifying key limitations, design constraints and potential gaps in research.
- Modelling of magnetic fields and their gradients from a ferromagnetic stent within a static field environment using COMSOL finite element analysis.
- Defining the magnetic field and gradient requirements for effective, intravascular cell capture under physiological flow.
- Modelling Poiseuille blood flow with Stokes drag at cerebral artery representative velocities using an analytical framework to validate COMSOL results under physiological flow.

- Quantifying capture range and efficiency across relevant SPION loadings and flows and validate the analytical framework against the COMSOL results.
- Document limitations and future work that would enhance insight and provide groundwork for experimental translation.

2 Research and Previous Work

2.1 Blood–Brain Barrier Constraints

The blood-brain barrier is formed of specialised endothelial cells with tight junctions, pericytes, basement membranes and astrocytic endfeet (glial cells) [9], illustrated in Figure 1, that provide the restrictive properties of the BBB.

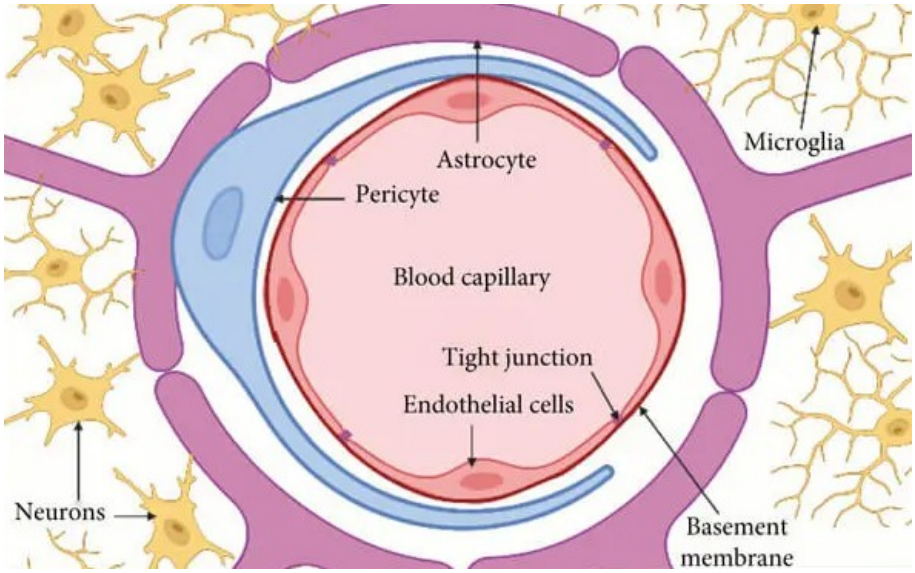


Figure 1: Schematic diagram of the BBB structure, illustrating endothelial cells, tight junctions, pericytes, basement membranes and astrocytic endfeet [10].

These restrictive properties are induced and maintained by critical interactions with mural cells, immune cells, glial cells and neural cells [3]. The tight junctions formed between adjacent cerebral endothelial cells significantly reduces paracellular diffusional pathways by providing size-selective permeability to uncharged molecules up to 4 nm and low permeability to larger molecules [11, 3]. This restriction in permeability is due to the presence of claudins, a membrane protein that binds homotypically on adjacent endothelial cells to form the primary seal of the tight junction [9, 11]. In particular, claudin-5 is the most expressed tight junction protein, governing the size-selective permeability [12]. Pericytes wrap around the endothelial cells, providing dynamic regulation through paracrine signalling which reduces leaky phenotypes and suppresses transcytosis along with leukocyte adhesion [11, 9]. The basement membranes act as the interface between the vasculature and the astrocytic endfeet, which cover approximately 99% of the cerebrovascular surface [13], controlling the movement of solutes between the tissues [14]. The astrocytic endfeet create a continuous glial sheath that modulates perfusion and maintains the BBB function via

neurovascular unit signalling to regulate nutrient supply, waste removal and blood flow [15, 11, 16]. This structure severely limits paracellular transport and transcytosis, which is necessary to protect the central nervous system from pathogens and toxins, ensuring ionic homeostasis required for neuronal excitability. However, this restrictive nature is the dominant obstacle to the delivery of cell therapeutics, meaning that current cell therapy to the CNS requires invasive, direct intracranial injection, which carries risk of haemorrhage and infection and most cells do not cross the CNS barriers [17].

2.2 Transport Mechanisms Across The BBB

The core mechanisms for transport across the BBB include: Receptor-Mediated Transcytosis (RMT), Adsorptive-Mediated Transcytosis (AMT), Carrier-Mediated Transport (CMT), paracellular transport and diffusion [18]. None of these methods work for cell-scale cargo without the disruption of the BBB. Whilst paracellular transport and diffusion are only viable during disease or inflammation of the BBB due to tight junction disruption [11, 3], transcytosis provides a method of large molecule transportation through endothelial cells via membrane-bound vesicles [19]. AMT allows for positively charged cargo including cationic proteins, peptides or modified antibodies to bind to the negatively charged endothelial surface and trigger endocytosis [20], the engulfing of membrane proteins with plasma membrane vesicles which occurs at the apical and basolateral membranes of the endothelial cells [21, 18]. RMT relies on ligand-receptor recognition to trigger endocytosis, allowing specific large molecules such as transferrin to cross the BBB [22] and provides a promising strategy for the delivery of drugs by engineering antibodies that mimic natural substrates of the receptors [18, 23]. RMT via transferrin or insulin receptors is the most established route for the transportation of large biologics such as IgG antibodies [24].

The utilisation of BBB disruption techniques such as Focused Ultrasound (FUS) can enable larger therapeutics such as stem cells to cross the BBB [6]. This method offers a non-invasive approach to increase BBB permeability by injecting gas-filled lipid shells intravenously and using FUS to create sustained, stable cavitation which produces local, reversible tight junction openings to increase endothelial transcytosis [25].

Whilst the BBB presents major obstacles in cell-based therapeutic delivery, the exploitation of selective transport methods such as RMT and BBB disruption techniques have enabled the entry of engineered molecules. However, the efficacy of drug and cell delivery through non-invasive measures remains an issue. The implementation of a magnetically active stent in a target cerebral vessel has the potential to enable local cellular retention, leading to increased dwell time near the vessel wall. This prolonged exposure could enhance the probability of transcytosis of engineered molecules or allow for cells to deliver therapeutic effects through paracrine signalling. Moreover, the advancement in BBB disruption techniques could be synergistically combined with the stent's retention model to create a controlled therapeutic gateway into the brain's nervous system.

2.3 Paracrine Signalling

Paracrine signalling is the secretion of bioactive molecules that act locally on neighbouring cells and is increasingly regarded as the dominant therapeutic route for MSC delivery [26, 27]. In studies of stroke models, MSCs are found to secrete vascular endothelial growth factor (VEGF), fibroblast growth factor-2 (FGF-2), brain-derived neurotrophic factor (BDNF) and various chemokines which drive angiogenesis (the formation of new blood vessels), suppress neuronal apoptosis (programmed cell death) and recruit endogenous progenitor cells (tissue-specific adult stem cells) [27, 28].

VEGF is an endothelial mitogen that induces a potent angiogenic response, mediated through the receptors VEGFR-1 and VEGFR-2 [29] and is the best-characterised factor. VEGF provides direct neuroprotective effects by enhancing neuronal survival and supporting neurogenesis [30] and has been shown to improve cognitive function in the hippocampus region, where it enhances the proliferation and survival of neural progenitor cells [31].

Ozawa et al. [32] identified that angiogenic outcome is governed by a threshold in microenvironmental VEGF production, rather than bulk concentration. Specifically, they found that VEGF doses of $5\,\mathrm{ng}/10^6$ cells/day to $70\,\mathrm{ng}/10^6$ cells/day produced stable angiogenesis, with levels above $100\,\mathrm{ng}/10^6$ cells/day resulting in aberrant, leaky vascularisation. The tissue half-life is approximately 60 min based on in vitro experiments [33, 34, 35], meaning that delivered cells that only survive a few hours or are swept away before establishing a local source population cannot produce a therapeutic VEGF field.

This strengthens the argument for a local device capable of cell retention, as even a modest capture efficiency is therapeutically meaningful if the captured cells are retained for long enough to provide a meaningful cytokine output that reaches the therapeutic threshold of $5\,\mathrm{ng}/10^6$ cells/day to $70\,\mathrm{ng}/10^6$ cells/day.

2.4 Magnetic Labelling and Targeting of Cells

Magnetic labelling of cells has allowed for locally targeted therapy by loading cells with biocompatible superparamagnetic iron oxide nanoparticles (SPIONs) such as magnetite (Fe_3O_4) [36]. These nanoparticles are internalised through direct culture loading, receptor-mediated uptake or transfection with magnetotactic-bacterial genes [37, 36]. Recent advancements in biomedical applications of magnetically enhanced cells have allowed for magnetic guiding of MSCs to sites of injury as discussed in the Sensenig et al. paper covering magnetic nanoparticle-based approaches to locally target therapy [38]. In this study they utilised externally applied magnetic fields to guide cells to target regions, however, noted that magnetic targeting is limited by depth of the target site and the presence of shear flow forces [38]. They also highlighted that a combination of high field and gradient strength is required to exert a force, but no single magnet optimises both. These limitations provide further argument for the development of an implantable system capable of enhancing magnetic gradients — the architecture discussed in this dissertation.

The study by Shen et al. on magnetic enhancement of stem-cell targeted delivery to the brain

during FUS also showed significant enhancement in delivery of neural progenitor cells in the brain that were labelled with SPIONs [6]. When a static magnetic field was applied over the target region, a greater number of SPION-labelled cells were present than non-labelled cells. However, as seen in previous studies, overall delivery efficiency was extremely low and in this case only around 0.03% of the injected dose was successfully delivered [6]. A novel transgenic approach has also been introduced by Elfick et al. whereby MSCs are transfected with the bacterial *mms6* gene from *Magnetospirillum magneticum* AMB-1, allowing for de novo intracellular synthesis of magnetic nanoparticles [37]. The engineered cells circumvent drawbacks from SPION loading including uptake variability, cytotoxicity and nanoparticle dilution after cell division, which can be utilised to further enhance long-term cell tracking and magnetic targeting [37].

MacDonald et al. demonstrated that applying a time-varying magnetic field can significantly increase the transport of magnetically loaded nanoparticles through viscous gels [39]. The addition of an oscillating component led to a 30-fold increase in particle transportation through the viscous gel at a magnetite loading of 30–50%. This indicates that time-varying fields may provide enhanced SPION transport under pulsatile flow conditions through dynamic field-flow coupling and is highlighted as a future implementation, as the present analysis uses steady Poiseuille flow as a first approximation under a uniform magnetic field.

Typical loadings of SPIONs are within 10–300 pg of iron oxide per cell, Polyak et al. [8] used 200 pg/cell for MSC labelling, the standard experimental upper end, which is used as a default reference in this analysis.

2.5 Magnetised and Functionalised Stents

Previous work has been carried out to overcome the limitations of external magnets through the use of implanted magnetic stents. Cell delivery is limited by shear forces and washout, so Pislaru et al. introduced stainless steel, chromium cobalt and nitinol stents to a 5000 G magnetic field, however, they did not retain any significant magnetic charge [40]. After further research, they found that nickel-coated stents showed significantly higher cell retention and found that the magnetically labelled endothelial cells migrated into the adjacent denuded vessel [40]. This process was carried out to facilitate vascular healing and has provided confirmation that the magnetisation of stents has practical medical application. However, the use of coatings can lead to leaching or corrosion over time, raising biocompatibility concerns, furthermore, increasing the thickness of the stent struts within a cerebral stent poses issues of deliverability and flow disruption.

Another approach focused on endothelial cell capture used ferromagnetic alloys for the stent itself. Tefft et al. developed a stent graft made from 2205 duplex stainless steel and used an external electromagnet to provide 5000 G of background magnetic field at the implant site [41]. The results showed that cells accumulated adjacent to the underlying struts, implying that capture is driven by strong local magnetic gradients rather than uniform surface attraction.

A computational model was also developed by Mardinoglu and Cregg to evaluate how the size of SPIONs affects capture in a stent-assisted magnetic drug targeting system [42]. The simula-

tion incorporated dipole-dipole forces, Stokes drag and hydrodynamic interactions to assess the capture efficiency and results showed that smaller SPIONs with higher saturation magnetisation produced stronger magnetic responses [42]. This study underscores the importance of optimising magnetic properties within cells to ensure maximum delivery efficiency and reinforces the feasibility of a cerebral stent being able to capture cells under physiological flow.

2.6 Target Regions

The success of the magnetically active cerebral implant relies heavily on its implant location within the cerebrovascular system as it will enforce design constraints and application use cases. Brain-Computer Interfaces rely on safe, intracerebral implantation and commonly use the venous sinuses such as the superior sagittal sinus due to its large diameter and low physiological flow conditions [43]. The transverse sinus is also attractive due to its large vessel diameter of 6.2–6.3 mm [44]. Whilst providing minimally invasive access via the jugular vein, as is done in the implantation of the Stentrode [43], the blood flow is directed away from the brain and is distanced from the brain microvasculature that this project intends to target.

For arterial options, the middle cerebral artery (MCA), anterior cerebral artery (ACA) and posterior cerebral artery (PCA) allow for the placement of the stent upstream of a target brain region, enabling systematic capture of cells as they flow past the retention site. The MCA supplies the lateral cerebral cortex, including the motor and sensory cortices, and is most commonly affected in ischaemic strokes [45]. As the MCA is the most commonly occluded intracranial artery, there is extensive research on endovascular thrombectomy within the MCA, making it a clinically validated route for neurointerventional stent delivery [46]. The ACA provides access to the medial frontal lobes and anterior corpus callosum, regions responsible for motor control, emotional regulation and decision-making [47], making it an area of interest in targeting neurological disorders. The PCA irrigates the occipital lobes and provides access to the thalamus and hippocampus [48], regions involved in the formation of memory and where damage can lead to amnesia [49].

These arterial vessels do have higher flow which poses significant challenges. However, the anchoring of the stent proximal to a downstream lesion enables indirect targeting, which mimics the logic already used in flow-diverting stents during aneurysm therapy [50].

3 Methodology

This section develops the analysis. Section 3.1 formulates the problem and requirements, Sections 3.2–3.4 develop the magnetostatics, including the geometry, Akoun-Yonnet analytical field, external field superposition, and the magnetic force with Langevin saturation. Sections 3.5–3.6 develop the hydrodynamics and the static capture criterion. Section 3.7 develops the trajectory ODE and Section 4.1 describes the COMSOL cross-validation.

3.1 Problem Formulation and Functional Requirements

Previous studies show that the development of a magnetisable stent for cell capture is feasible and can provide useful applications in targeted therapeutic delivery. The problem addressed by this project is the analysis of an implantable magnetic retention device, capable of capturing SPION-labelled cells under physiological cerebral flow conditions.

The analysis assesses the effective capture range and efficiency against multiple variable parameters. The target vessel selected is the MCA, which is modelled as a straight cylindrical tube of inner radius $R_{\text{ves}} = 1.5\,\text{mm}$. The default mean velocity used is $\bar{v} = 0.2\,\text{m/s}$, which is a conservative representation of haemodynamically compromised MCA flow, with parametric sweeps extending the velocity range to$\bar{v} = 0.2\,\text{m/s}$, representative of healthy flow [51]. The blood is modelled as a Newtonian fluid with dynamic viscosity $\eta = 4\,\text{mPa}\,\text{s}$ and density $\rho = 1060\,\text{kg/m}^3$ in steady Poiseuille flow, justified by a Womersley number $\alpha \approx 2$, placing the flow in the quasi-steady regime. COMSOL v6.1 is used as a finite element model to map the magnetic gradient and field strength from the struts of the stent, which were designed using CAD. The magnetic field imposed on the stent is a $B_0 = 1.5\,\text{T}$ uniform field acting along the z-axis, representing a clinical MRI-strength field.

The analytical framework is then used to perform parametric sweeps and assess the feasibility of the model under physiological flow. The analytical framework models the stent as a single planar ring of radially magnetised rectangular struts, isolating the dominant capture physics without loss of generality. The magnetic field is computed analytically using the Akoun-Yonnet closed-form expression [52] for a uniformly magnetised rectangular prism, which is superposed across the struts with the same uniform $B_0 = 1.5\,\text{T}$ field applied along the z-axis as in the COMSOL analysis. The Furlani–Ng scalar-gradient model [53] is utilised with a Langevin saturation correction [7] and Stokes hydrodynamic drag [54] to model the forces experienced by the SPION-loaded cells. Cell capture is defined as occurring when F_{mag} overcomes F_{drag} and deflects the cell onto the stent surface over the 2 mm upstream approach length. The fixed parameters used throughout the analysis are listed in Table 2.

Throughout the analysis multiple parametric sweeps are carried out to define capture performance across the operating envelope, these are discussed in the relevant methodology and results sections.

3.2 Stent Geometry

3.2.1 CAD Models

The COMSOL analysis was performed on three-dimensional stent CAD models imported as STEP files. Four stent geometries were used in the analysis. These models follow a braided closed-cell architecture, Figure 2, in which diamond-shaped cells are arranged circumferentially around a cylindrical tube of radius $R = 1.5\,\text{mm}$, sized to the inner wall of the M1 MCA [55]. The models all have the same strut width and thickness as stated in Table 2, and vary in the number of diamond-cells to assess the impact of stent porosity and the number of strut junctions

Table 2: Fixed parameters defining the physical baseline.

Category	**Parameter**	**Value**
Vessel	R_{ves}	1.54 mm
	$\bar{v}$	0.2 m/s
	η	4 mPa s
	ρ	1060 kg/m^3
Stent geometry	R	1.5 mm
	w	100 µm
	t	80 µm
	L	500 µm
	Magnetisation	Radial
Cells	R_{cell}	10 µm
	χ_0	2.0
	M_s	446 kA/m
	ρ_{SPION}	5170 kg/m^3
External field direction		Axial ($+z$)
Trajectory domain	Approach length	2 mm
	Integrator	RK45
	Tolerances	rtol $= 10^{-6}$, atol $= 10^{-9}$

at the mid-plane cross-section.

Fillets of radius 0.03 mm were applied at all strut junctions to improve meshing quality and reflect fabrication constraints, although meshing artefacts still occur and are discussed in section 4.1.

3.2.2 Analytical Model

The braided stent geometry is approximated by a single planar ring of n uniformly magnetised rectangular struts for three reasons. Firstly, this provides the means for efficient parametric sweeps without multiple, computationally expensive, simulation runs whilst still representing the magnetostatic equivalent of the braided geometry at an axial cut plane. Secondly, the contribution to the gradient field from axial strut segments accounts for a fraction of the local gradient magnitude at capture distances of 0–200 µm, with a fitted power-law exponent $n \approx 1.3$–1.4 (Section 4.1.3). Finally, the geometric correction that accounts for the oblique braid angle is accounted for by the calibrated effective magnetisation $M = 2.20$ MA/m, which was fitted to the gradient decay data from COMSOL, providing agreement to within 1% between the two stent geometry models (Section 4.1.4).

Each strut in the analytical model is a rectangular prism of width $w = 100$ µm, thickness $t = 80$ µm, and length $L = 500$ µm. The struts are equally spaced at azimuths $\theta_i = 2\pi i/n$ on a ring of radius $R = 1.5$ mm. The lumen available for cell transport has inner radius $r_{\text{inner}} = R - t/2 = 1.46$ mm, with the vessel wall at $R_{\text{ves}} = 1.54$ mm. This representative model isolates the dominant capture physics without loss of generality and allows for dense parametric sweeps as presented in Section 4.

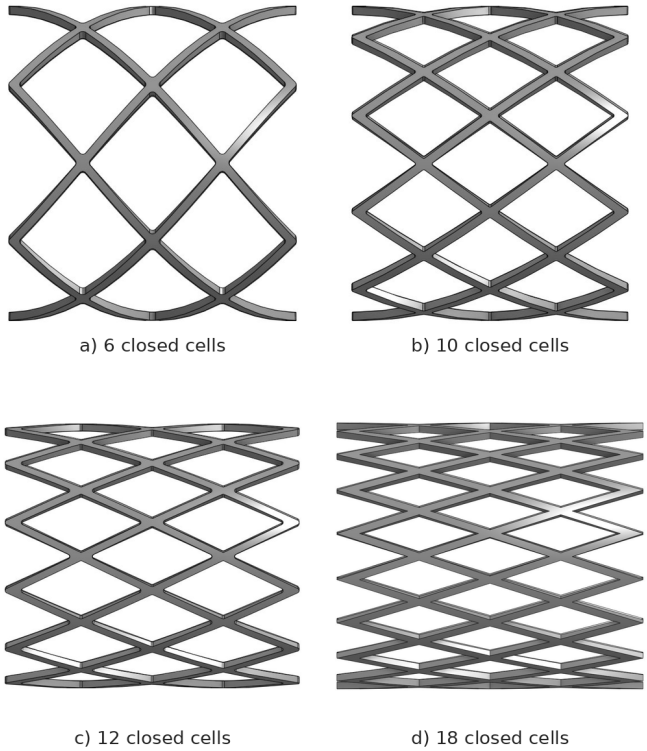


Figure 2: Stent geometry side view: a) V1 – 6 closed cells, b) V4 – 10 closed cells, c) V2 – 12 closed cells, d) V3 – 18 closed cells.

3.3 Magnetic Field Model

3.3.1 Akoun-Yonnet Analytical Field

Akoun and Yonnet [52] derived a closed-form analytical expression for the magnetic flux density of a uniformly magnetised rectangular prism — the governing geometry used to represent each strut. The derivation treats each rectangular prism as a distribution of surface magnetic charge density $\sigma_m = M\hat{n}$ over the six faces, where $\hat{n}$ is the outward surface normal. These surface charges integrate to a sum of arctan and logarithmic terms at the eight corners of the prism.

Since there are no free currents outside the prism, $\nabla \times \mathbf{H} = 0$ in the lumen, where $\mathbf{H}$ is the magnetic field strength and relates to the magnetic flux density $\mathbf{B}$ through the relation:

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \tag{1}$$

Where μ_0 is the permeability of free space and $\mathbf{M}$ is the magnetisation of the medium. Outside the strut material, $\mathbf{M} = 0$ and this simplifies to $\mathbf{B} = \mu_0\mathbf{H}$, so the lumen field can be written as the gradient of a scalar magnetic potential:

$$\mathbf{H} = -\nabla\phi_m, \qquad \phi_m(\mathbf{r}) = \frac{1}{4\pi}\int_S \frac{\sigma_m(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\,\mathrm{d}S' \tag{2}$$

For a rectangular face with uniform σ_m, this reduces to a double integral of $1/R$ over the face area, where $R = \sqrt{U^2 + V^2 + W^2}$ and (U, V, W) are the coordinates of the observation point measured relative to a corner of the prism. The antiderivative is the closed-form function:

$$F(U, V, W) = U \ln(V + R) + V \ln(U + R) - W \arctan\left(\frac{UV}{WR}\right) \tag{3}$$

Summing this function across the eight corners of the prism:

$$\phi_m \propto \sum_{i,j,k \in \{0,1\}} (-1)^{i+j+k} F(U_{ijk}, V_{ijk}, W_{ijk}) \tag{4}$$

with $\mathbf{H} = -\nabla\phi_m$ gives the field components in the local prism frame. For a prism of half-dimensions ($a = t/2,\ b = w/2,\ c = L/2$) magnetised along its local $+x$ axis, only the two faces perpendicular to $\mathbf{M}$ carry surface charge and contribute to the sum, the remaining four faces have $\mathbf{M} \cdot \hat{n} = 0$ and contribute nothing.

Each strut is evaluated in its own local frame and the resulting field vector is rotated back to the global frame by:

$$\mathbf{r}_i' = \mathbf{R}_i^T \mathbf{r}, \qquad \mathbf{B}_i^{\text{global}} = \mathbf{R}_i \, \mathbf{B}_i^{\text{local}} \tag{5}$$

where $\mathbf{R}_i$ is the 3×3 rotation matrix that maps the local $+x'$ axis of strut i onto the global radial direction at azimuth θ_i. The magnetisation direction was set to radial (local $+x$) for all parametric studies, as it maximises the radial component of the magnetic gradient towards the lumen, consistent with diametric magnetisation using an external magnet [41]. The total stent field is then given by superposition over all n struts:

$$\mathbf{B}_{\text{stent}}(\mathbf{r}) = \sum_{i=1}^{n} \mathbf{R}_i \, \mathbf{B}_i^{\text{local}}\left(\mathbf{R}_i^T \, \mathbf{r}\right) \tag{6}$$

This superposition relies on the assumption that the magnetisation of each strut is prescribed and not altered by the field of its neighbours. This is the same assumption that the COMSOL calibration corrects for, as discussed in Section 3.3.3.

3.3.2 External Field and Superposition

A uniform external field $B_0 = 1.5\,\text{T}$, simulating MRI conditions, is superposed on the stent field, making the total field at any point $\mathbf{B}_{\text{total}} = \mathbf{B}_{\text{stent}} + \mathbf{B}_0$. However, when computing the gradient of the total field magnitude, $\nabla|\mathbf{B}_{\text{tot}}| \neq \nabla|\mathbf{B}_{\text{stent}}|$ even though $\nabla\mathbf{B}_0 = 0$, because the gradient of a vector magnitude is non-linear in the constituent fields. This causes the total field to, in regions where $\mathbf{B}_0$ does not align with $\mathbf{B}_{\text{stent}}$, rotate away from the radial direction, which reduces the projection of $\nabla\mathbf{B}_{\text{stent}}$, thereby suppressing $\nabla|\mathbf{B}_{\text{tot}}|$ relative to $\nabla|\mathbf{B}_{\text{stent}}|$ alone. Even though the total gradient has been suppressed, the physically correct capture metric is the force

parameter $F_p = |\mathbf{B}_{\text{tot}}| \cdot \nabla |\mathbf{B}_{\text{tot}}|$ (units: T^2/m). The quantified effect of B_0 on the force parameter is presented in Section 4.2.1.

3.3.3 COMSOL Field and Calibration to Analytical Model

The stent material is modelled in COMSOL as a linear, soft ferromagnetic steel with relative permeability $\mu_r = 2$. This value is not a direct material measurement but was adopted to represent the material response at magnetic saturation, as ferromagnetic steels exhibit high initial permeability at low fields which decreases substantially as the applied field approaches saturation [56], which is true for MRI-relevant field strengths. The surrounding domain was spherical with a radius of 10 mm, sufficient to render boundary effects negligible, and used $\mu_r = 1$ to represent the diamagnetic nature of blood.

The background uniform magnetic field used the COMSOL `AC/DC -- Magnetic Fields No Currents` module (`mfnc`) and was set to 1 193 662 A/m along the z-axis, representative of a 1.5 T field through the constitutive relation in free space, $\mathbf{B} = \mu_0 \mathbf{H}$. Free-triangular meshing was used with the 'extremely fine' grid to remove mesh singularities at strut corners. Zero magnetic scalar potential ($\phi_m = 0$) was imposed on all outer boundaries of the spherical domain, valid due to the negligible amplitude of the stent perturbation field at the boundaries.

The analytical model uses a prescribed magnetisation M for the struts, an approximation that does not include the material permeability or the response of the stent to the applied field. However, COMSOL solves the full magnetostatic boundary value problem self-consistently, computing the induced magnetisation as a function of the local field everywhere in the domain. Therefore, the Akoun-Yonnet analytical model requires a calibration step to find the value of M that results in a gradient profile matching the COMSOL solution in the physiologically relevant capture zone before parametric sweeps are performed.

The calibration used the field condition $B_0 = 1.5\,\text{T}$, which drives the stent material into the saturation region, where the stent acts as a flux concentrator. The calibrated value $M_{\text{eff},1.5\,\text{T}} = 2.20\,\text{MA/m}$ accurately reproduces both the 100 T/m crossing ($d = 0.240\,\text{mm}$) and the 40 T/m crossing ($d = 0.380\,\text{mm}$) to within 1% of the gradient data values of the V2 stent design. All of the results obtained use the calibrated $M_{\text{eff},1.5\,\text{T}} = 2.20\,\text{MA/m}$ for cross validation to the COMSOL results.

3.3.4 Field Gradient Computation

Within COMSOL, the magnetic gradient field was computed using the equation:

$$\sqrt{\left(\frac{\partial\,\texttt{mfnc.normB}}{\partial x}\right)^2 + \left(\frac{\partial\,\texttt{mfnc.normB}}{\partial y}\right)^2 + \left(\frac{\partial\,\texttt{mfnc.normB}}{\partial z}\right)^2} \tag{7}$$

where the partial derivatives denote the numerical differentiation of the scalar field `mfnc.normB` on the mesh computed by COMSOL's built-in finite-difference scheme. The solution was mapped onto a 2D cut-plane at $z = 0$, which passed through the strut junctions where the gradient

magnitude is at a maximum (Section 4.1.2). The gradient profile was extracted by using a 3D cut-line from the inside of the strut junction at $r = 1.42\,\text{mm}$, acting radially inwards towards the vessel centreline along the azimuth $\theta = 0$. This data was then exported to a `.csv` file and cleansed by removing non-physical discontinuities caused by numerical artefacts at element boundaries. Mesh independence was verified by repeating the gradient extraction after halving the element count, arbitrary threshold-crossing distances changed by less than approximately 10 µm, confirming the data is mesh-converged in the relevant capture zones.

For the analytical module, gradients were computed numerically using central finite differences with a step size of $\delta = 500\,\text{nm}$, applied to the field at the interface of the relevant domain (stent only, or total field including $\mathbf{B}_0$). The magnitude of the field gradient is given by:

$$\nabla|\mathbf{B}|(\mathbf{r}) \approx \frac{|\mathbf{B}(\mathbf{r} + \delta\,\hat{e}_k)| - |\mathbf{B}(\mathbf{r} - \delta\,\hat{e}_k)|}{2\delta}\,\hat{e}_k, \quad k = x, y, z \tag{8}$$

This is a standard second-order central difference approximation. The truncation error is $\mathcal{O}(\delta^2)$, which is confirmed by a convergence check, whereby halving the step size reduced the gradient error quadratically, see Appendix A, Fig. A.1.

3.4 SPION-Labelled Cell Model

3.4.1 Parameters and Saturation Model

SPION-labelled MSCs circulating in blood flow were modelled as non-rotating spheres of radius $R_{\text{cell}} = 10\,\mu\text{m}$. The internalised SPION volume within the cell is given by:

$$V_{\text{SPION}} = \frac{m_{\text{SPION}}}{\rho_{\text{SPION}}} \tag{9}$$

The SPION-labelled cell parameters are displayed in Table 3. The default SPION loading used is 200 pg per cell, the experimental optimal loading used by Polyak et al. [8], with parametric sweeps over the range 1–300 pg/cell, spanning literature values from Riegler et al., Polyak et al. and Mardinoglu et al. [57, 8, 42].

Table 3: SPION-labelled cell parameters.

Parameter	Symbol	Value	Unit	Reference
Cell radius	R_{cell}	10	µm	[58]
SPION mass per cell (default)	m_{SPION}	1–300	pg	[8]
SPION volume susceptibility	χ_{SPION}	2.0	—	[53]
SPION material density (magnetite)	ρ_{SPION}	5170	kg/m^3	[53]
SPION saturation magnetisation (bulk magnetite)	M_{sat}	446×10^3	A/m	[53]

$R_{\text{cell}} = 10\,\mu\text{m}$ was chosen to represent the lower end of cultured MSC size distributions (15–30 µm) and is representative of the small-sized MSC population selected for intra-arterial delivery to minimise vascular obstruction [58].

The SPION susceptibility is treated as field-dependent to account for saturation near the stent surface. Magnetite SPIONs exhibit a saturation magnetisation of 4.78×10^5 A/m [53], corresponding to an equivalent field scale of roughly 0.6 T, meaning that at $B_0 = 1.5$ T the particles are fully saturated. A Langevin correction is therefore implemented to calculate an effective susceptibility:

$$\chi_{\text{eff}}(B) = \chi_0 \, f_{\text{sat}}(B), \quad f_{\text{sat}}(\xi) = \frac{3\mathcal{L}(\xi)}{\xi}, \quad \mathcal{L}(\xi) = \coth(\xi) - \frac{1}{\xi} \tag{10}$$

With the dimensionless field parameter $\xi = \alpha B/\mu_0$ and $\alpha = 3\chi_0/M_{\text{sat}}$. At low fields ($\xi \to 0$), $f_{\text{sat}} \to 1$ and $\chi_{\text{eff}} \to \chi_0$ (linear regime). At high fields, SPIONs become saturated and the χ_{eff} decreases as $f_{\text{sat}} \to 3/\xi \to 0$. A Taylor expansion $\chi_{\text{eff}} \approx \chi_0 \left(1 - \xi^2/15\right)$ was used for numerical stability at small $\xi < 10^{-3}$.

3.4.2 Magnetic Force on a Cell

The gradient force expression derived by Furlani and Ng [53] was used to calculate the magnetic force action on the SPION-labelled cells, which for a spherical particle with volume susceptibility χ_{SPION} is given by:

$$\mathbf{F}_{\text{mag}} = \frac{V_{\text{SPION}} \, \chi_{\text{eff}}}{\mu_0} \, |\mathbf{B}_{\text{total}}| \, \nabla |\mathbf{B}_{\text{total}}| \tag{11}$$

The direction of $\mathbf{F}_{\text{mag}}$ follows $\nabla|\mathbf{B}_{\text{tot}}|$, pointing towards the stent strut surface from along increasing field magnitudes.

3.5 Hydrodynamics

3.5.1 Blood Flow Model

Blood flow is modelled as steady, laminar, Poiseuille flow in a straight cylindrical vessel, with the axial velocity profile given by:

$$v_z(r) = 2\bar{v}\left(1 - \left(\frac{r}{R_{\text{ves}}}\right)^2\right), \quad r = \sqrt{x^2 + y^2} \tag{12}$$

Where $\bar{v}$ is the mean velocity across the cross section and is swept through 0.05 m/s to 0.6 m/s, spanning diastolic, diseased-vessel flow to fast, healthy MCA representative flows [51]. Peak velocity occurs at the vessel centreline where $v_{\text{max}} = 2\bar{v}$. Poiseuille flow was chosen as a simplified model of pulsatile flow, justified by a Womersley number of $\alpha \approx 2$ for physiological pulsatility at 1 Hz:

$$Wo = R\sqrt{\frac{\omega\rho}{\mu}} \tag{13}$$

Which places the flow in the quasi-steady regime due to the oscillatory boundary layer thickness being, $\delta \approx R/\alpha \approx 0.75mm$, which is comparable to the vessel radius. Poiseuille flow is therefore a first-order approximation for time-averaged capture analysis and the implementation of a pulsatile profile which is discussed in the limitations section.

3.5.2 Stokes Drag

The viscous drag acting on a spherical cell moving at velocity v_{cell} through blood at local velocity v_{blood} is calculated using the Stokes relation:

$$\mathbf{F}_{\text{drag}} = 6\pi\eta R_{\text{cell}}(\mathbf{v}_{\text{blood}} - \mathbf{v}_{\text{cell}}) \tag{14}$$

The Reynolds number for a cell of radius 10 µm at 0.2 m/s is $Re \approx 1$, which is at the upper end limit of Stokes validity, however, the radial component of the cell's velocity towards the vessel wall is comfortably within the Stokes regime and so Stokes relation is an acceptable approximation. See Appendix A, Fig. A.2 for the full drag characterisation.

3.6 Static Capture Criterion

The scalar criterion of Furlani and Ng [53] considers a cell capturable if the magnitude of magnetic force experienced by the cell exceeds the hydrodynamic drag force: $|F_{\text{mag}}(\mathbf{r})| > |F_{\text{drag}}(\mathbf{r})|$. This conservative approach however neglects the directionality of the forces, as F_{drag} is primarily axial and F_{mag} is primarily radial. It also evaluates drag at the full local velocity which does not account for any radial drift the cell has accrued over the approach distance.

The static capture range is defined as the maximum distance from the struts inner surface, along a radial inward sweep, where the capture criterion is satisfied. This analytical model used 400, equally spaced, evaluation points along the strut-radial line. At representative MCA conditions, F_{mag} is much lower than F_{drag}, so the criterion is only satisfied within a thin boundary layer, significantly underestimating the capture range compared to the trajectory-integration model (Section 4.2.4).

3.7 Trajectory Integration

The trajectory model for a SPION-labelled cell solves Newton's second law in the low Reynolds number regime, where inertia is negligible and the cell velocity is given by the first-order ODE:

$$\frac{d\mathbf{r}}{dt} = \mathbf{v}_{\text{blood}}(\mathbf{r}) + \frac{\mathbf{F}_{\text{mag}}(\mathbf{r})}{6\pi\eta R_{\text{cell}}} \tag{15}$$

The cell is always at terminal velocity because the viscous relaxation time is five orders of magnitude below the force-change timescale:

$$T_{\text{viscous}} = \frac{m_{\text{cell}}}{6\pi\eta R_{\text{cell}}} \approx 1\,\mu\text{s} \tag{16}$$

Equation (15) was solved using the Runge-Kutta RK45 method via `scipy.integrate.solve_ivp`, with tolerances rtol = 10^{-6} and atol = 10^{-9}. Three terminal events are used to classify cell outcomes:

1 — When z reaches $z_{\text{end}} = +2\,\text{mm}$ downstream: the cell is classified as *escaped*.

2 — When the cell's transverse radius $\sqrt{X^2 + Y^2}$ reaches $R - t/2$, the stent inner surface: the cell is classified as *captured*.

3 — When a cell approaches a strut from outside the ring, signalling a strut-proximity event: the cell is classified as *captured*.

The integration window begins at $z = -2\,\text{mm}$ and terminates on the first event. The capture range is obtained through a 7-step binary search over the initial radial injection position, giving $\approx 11\,\mu\text{m}$ resolution.

4 Results

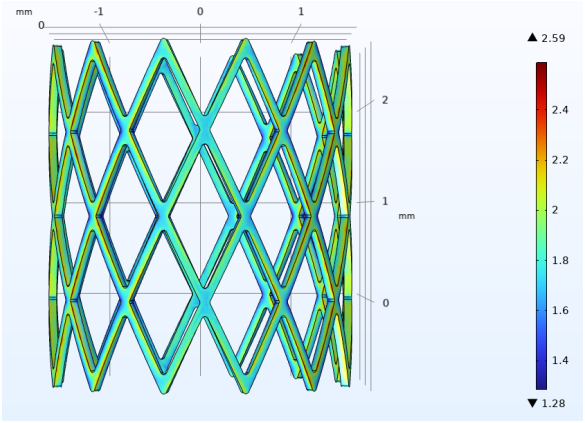
4.1 COMSOL Finite-Element Simulations

Four three-dimensional CAD models, created in Onshape, were imported into COMSOL Multiphysics v6.1 using the Magnetic Fields, No Currents (`mfnc`) physics module as described in Section 3.3.3. The four models — V1 ($N = 6$), V2 ($N = 12$), V3 ($N = 18$) and V4 ($N = 10$) — represent the corresponding braided CAD geometries (Section 3.2.1). The V2 model was used as the primary dataset for analysis, with the other models used to support key findings. COMSOL was used to evaluate and visualise the magnetic flux density $|\mathbf{B}|$ (`mfnc.normB`) across the stent model surface, and the magnetic flux gradient $\nabla|\mathbf{B}|$, represented by Equation (7), across a cut-plane at $z = 0$, where strut junctions produce maximum gradient magnitude. The stent material was assigned a permeability value of $\mu_r = 2$, representative of saturated soft-ferromagnetic material, and a uniform external field of $B_0 = 1.5\,\text{T}$ was imposed along the z-axis. All gradient values reported were obtained from 3D cut-line exports from the strut junction orthogonal to the field direction, extending radially towards the vessel centreline. Near-surface singularity points were excluded with model-specific cut-off values as follows: $d > 25\,\mu\text{m}$ for V1 (to exclude a non-monotone corner cluster), $d > 5\,\mu\text{m}$ for V2 and V3, and the V4 dataset restricted to $d < 170\,\mu\text{m}$ to exclude secondary-strut interference artefacts in the mid-lumen.

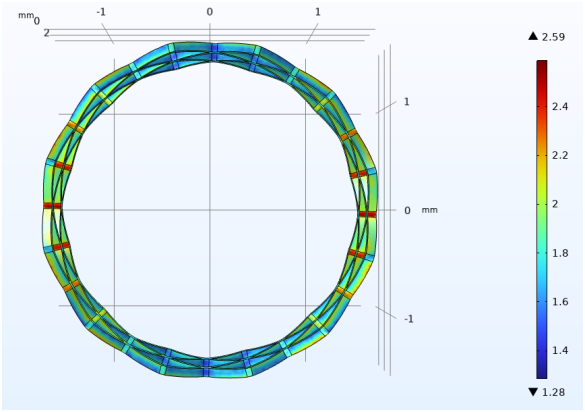
4.1.1 Magnetic Flux Density Distribution

Figure 3 shows the `mfnc.normB` surface plots for the V2 stent geometry across four planes. This figure demonstrates that field enhancement is localised at strut junction corners, where peak values approach $\approx 2.3\,\text{T}$ to $2.5\,\text{T}$, approximately $0.8\,\text{T}$ to $1.0\,\text{T}$ above the background field $|\mathbf{B}_0|$. This concentration in flux is a consequence of flux channelling through the stent struts,

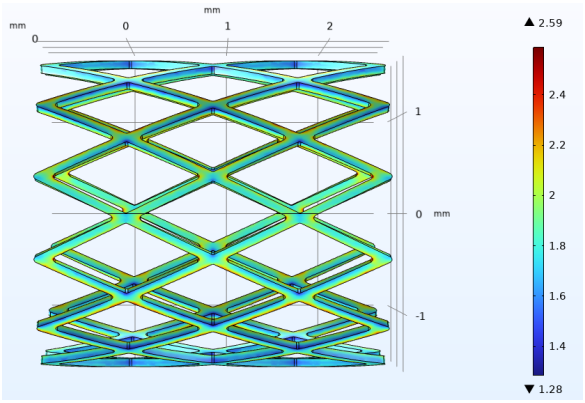
with struts aligned axially with the background field yielding higher flux density distributions. This is governed by magnetic reluctance and the geometry-dependent demagnetisation factor $\mathbf{N}$ of each stent strut [59], which accounts for the shape-induced opposing field produced by surface magnetic poles, reducing the effective field $\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{app}} - \mathbf{NM}$. The magnitude of this opposing field depends on the alignment of the body's dimensions along the field direction. Struts orientated along the z-axis present a small demagnetisation factor, whereas struts orientated perpendicular to $\mathbf{B}_0$ have a higher demagnetisation factor. The external field also follows a path of least resistance, meaning that preferential channelling further enhances flux concentration in axially aligned struts. However, struts aligned perpendicularly to $|\mathbf{B}_0|$ force the field lines to pass through the narrow cross-sectional faces of the strut, meaning that the flux density remains close to $|\mathbf{B}_0|$ and in some areas decreasing to $|\mathbf{B}| \approx 1.3\,\text{T}$ to $1.4\,\text{T}$, as visualised by the dark blue sections on struts aligned with the x-axis in Figure 3. The implication of this for SPION capture is that the gradient magnitude (and hence magnetic retention force) is strongest near axially aligned struts.



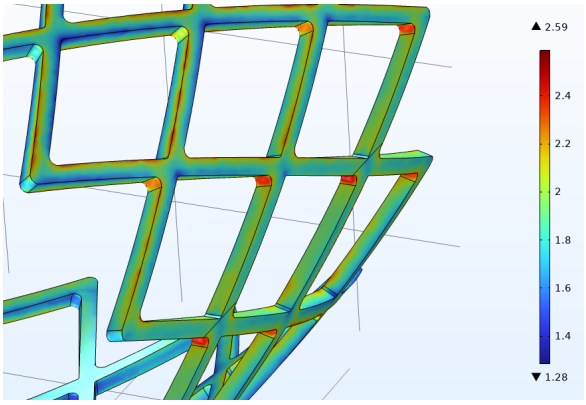
(a) xy-plane ($z = 0$): azimuthal distribution.



(b) xz-plane: axial profile through one strut.



(c) zy-plane: cross-sectional view.



(d) Zoomed diagonal cut: near-strut flux concentration.

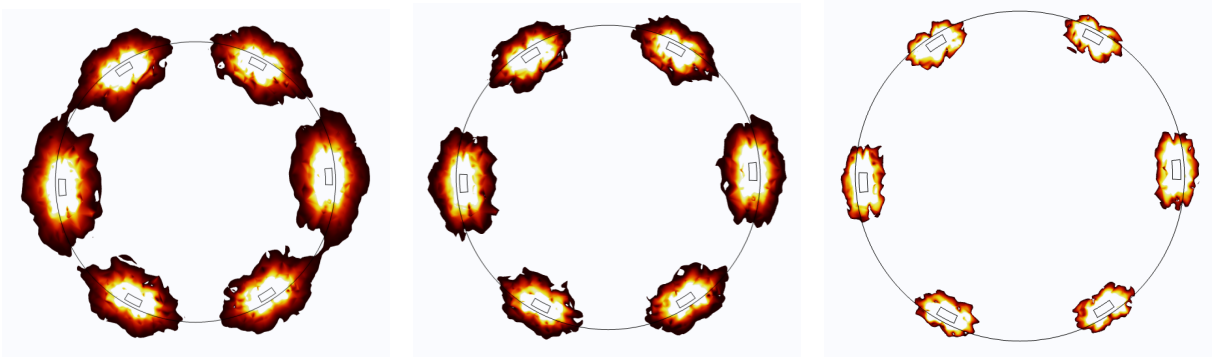
Figure 3: Magnetic flux density magnitude `normB` for V2 ($N = 12$) at $B_{\text{ext}} = 1.5\,\text{T}$. Colour scale in mT.

The same magnetostatic properties of magnetic flux density concentrations were observed across all other models.

4.1.2 Magnetic Gradient Heatmaps and Threshold Contours

The magnetic flux gradient magnitude $|\nabla|\mathbf{B}||$ was calculated using Equation (7) and plotted at the $z = 0$ mid-plane across all geometries, as shown in Figures 4–6. Three gradient thresholds are used throughout as spatial reference levels for characterising gradient decay: 40 T/m, 100 T/m and 300 T/m. The 40 T/m value originated from an initial order-of-magnitude force balance estimate early in the study, prior to the full trajectory analysis implementation, and were carried forward for consistency across figures, with 100 T/m and 300 T/m selected as representative markers for comparison. These values are not derived at MCA representative conditions and should be treated purely as spatial reference markers for gradient decay and COMSOL cross-model validation.

In the V1 model the six capture zones are tightly confined around each strut with no inter-strut merging across the full gradient range (Figure 4) due to the inter-strut arc being too large relative to the field decay. In the V2 and V3 models, there is partial superposition of adjacent strut fields which produces an enclosed zone of zero gradient, as seen in Figure 5. This phenomenon occurs as a result of $\nabla|\mathbf{B}|$ being a non-linear function of the constituent field vectors. When the two gradient fields merge, constructive interference ridges are formed due to the radial components of the gradient field vectors reinforcing one another, when the tangential components from each strut are orientated in opposing directions they partially cancel, smoothing local $|\mathbf{B}_{\text{tot}}|$ and collapsing the scalar gradient magnitude $\nabla|\mathbf{B}|$, forming a null gradient region. A SPION-labelled cell entering this region will experience negligible capture force, and this represents an intrinsic limitation of any discrete multi-strut ring geometry. Increasing the gradient threshold to 300 T/m eliminates the constructive inter-strut interference and shows that the gradient field is confined around the struts (Figure 5) as in V1.

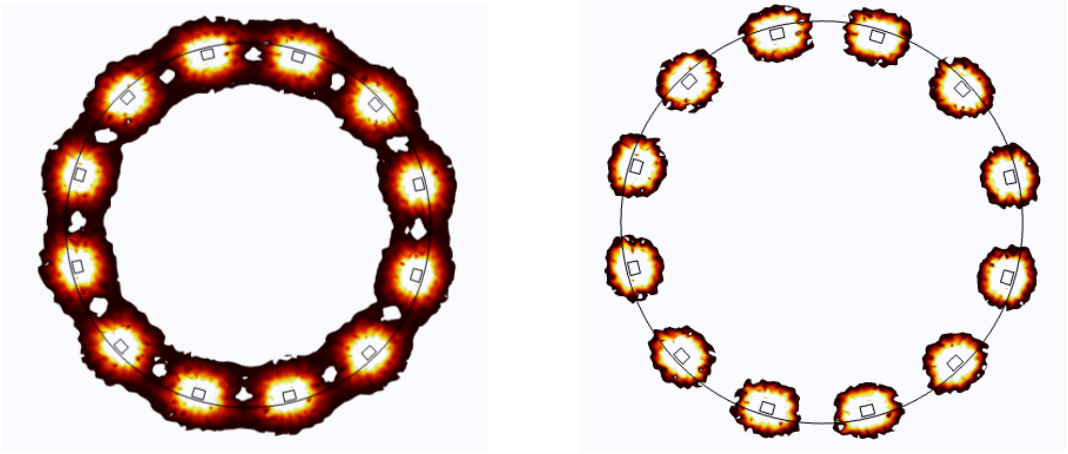


(a) 40 T/m (b) 100 T/m (c) 300 T/m

Figure 4: V1 model $|\nabla|\mathbf{B}||$ heatmaps at the $z = 0$ mid-plane for threshold contours of (a) 40 T/m, (b) 100 T/m, and (c) 300 T/m. Capture zones are tightly confined around each individual strut with no inter-strut merging across the full gradient range.

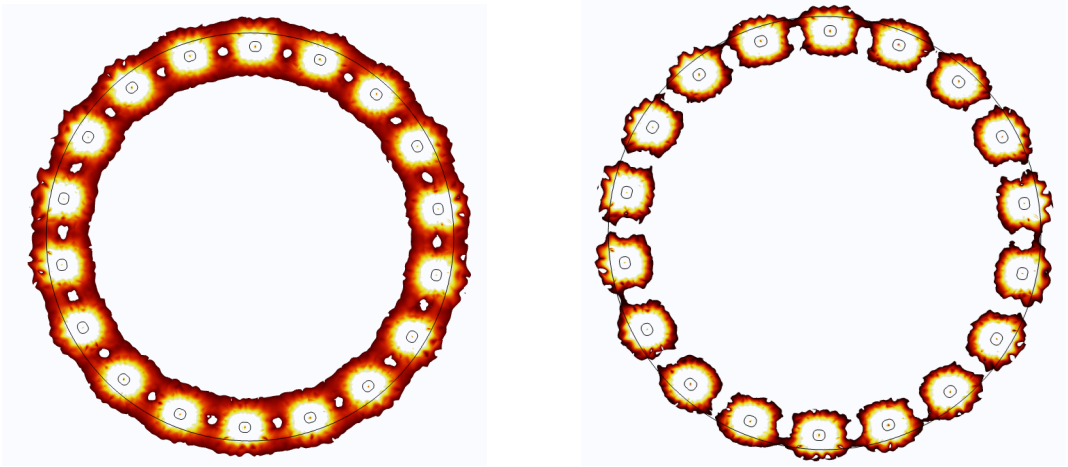
4.1.3 Radial Gradient Decay Profiles and Power-Law Fits

The radial gradient decay data was created using 3D cut-line profiles, originating at a stent strut aligned axially with the external field and extending radially towards the vessel centreline.



(a) 100 T/m (b) 300 T/m

Figure 5: V2 model $|\nabla|\mathbf{B}||$ heatmaps at the $z = 0$ mid-plane. (a) At 100 T/m the null region is clearly enclosed by constructive interference ridges. (b) At 300 T/m the inter-strut interference is eliminated and capture zones are confined to the immediate strut vicinity.



(a) 100 T/m (b) 300 T/m

Figure 6: V3 model $|\nabla|\mathbf{B}||$ cut-plane heatmaps at threshold contours of (a) 100 T/m and (b) 300 T/m. Inter-strut superposition behaviour is consistent with V2. Note: the lighter red colour was caused by an error in contour mode selection but still represents the 100 T/m boundary.

The data was cleansed by removing data points where $d < 25\,\mu\text{m}$, as strut surface singularities, field transitions from the ferromagnetic domain to free space and numerical gradient evaluations were unreliable at this range. A power-law fit was then applied:

$$G(d) = A \cdot d^n, \qquad d > 0 \tag{17}$$

where d is the arc length from the strut surface in mm, G is $\nabla|\mathbf{B}|$ in T/m, A is the gradient value at $d = 1\,\text{mm}$, and A_{100} is the gradient value at $d = 100\,\mu\text{m}$. This power-law fit is motivated by the far-field dipole limit of the Akoun-Yonnet field expression [52] where the field decays as r^{-3}. The actual decay does not match r^{-3} as the physiologically relevant observation range (25 µm to 300 µm) does not satisfy the far-field criterion, which is why the power law is fitted to the COMSOL data. The fit was performed using `scipy.curve_fit` in linear space, which implements the Levenberg-Marquardt algorithm that iteratively minimises the sum of squared residuals [60]. The fitted parameters are summarised in Table 4.

Table 4: Power-law fit parameters $G(d) = A \cdot d^n$ for each COMSOL model. Data points at $d < 25\,\mu\text{m}$ excluded from all fits.

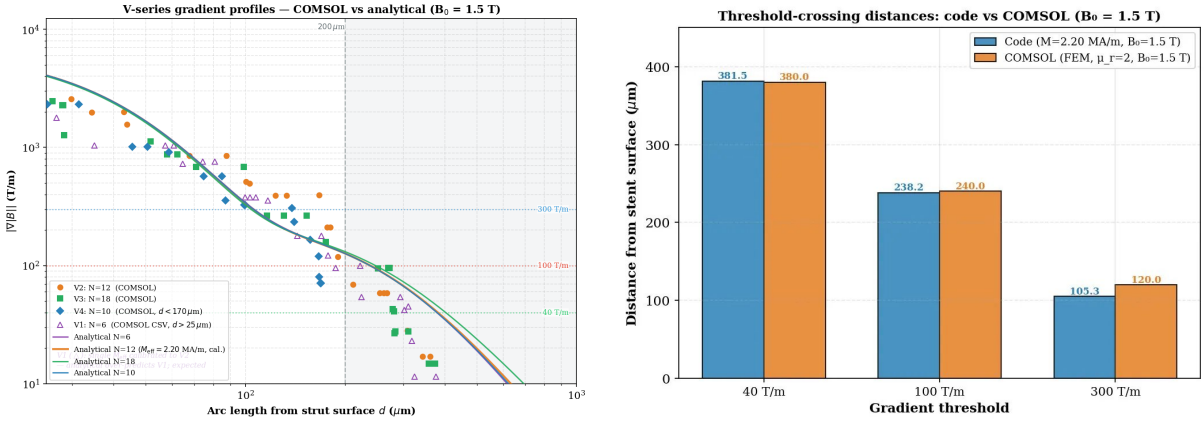
Model	N	A (T/m/mm^n)	n	R^2	n_{pts}	A_{100} (T/m)
V1	6	24.75	−1.247	0.956	41	437
V2	12	23.77	−1.340	0.976	38	520
V3	18	23.47	−1.253	0.957	38	420
V4	10	23.79	−1.241	0.950	18	414

Two conclusions can be drawn from the power-law fits in Table 4. Firstly, the decay exponents $n = -1.241$ to -1.340 are consistent within $\approx 7.7\%$, which implies that the strut count has minimal influence on the functional form of gradient decay within the braided ring geometry used across the models. Secondly, the near-surface gradient evaluated at $d = 100\,\mu\text{m}$ is within a factor of 1.256 across all models, which with fit uncertainties of 22–36% at A_{100} renders the differences in A_{100} values indistinguishable. This further solidifies the conclusion that the capture-relevant gradient acting radially towards the vessel centreline near the stent strut is primarily defined by the width, thickness and magnetisation of the strut rather than strut count. This also supports the phenomenon observed in Section 4.1.2 that strut count governs the azimuthal coverage of the gradient field, not the gradient magnitude available for capture along the 3D cut-lines.

4.1.4 Analytical Cross-Validation

The COMSOL solution for the V2 model ($B_0 = 1.5\,\text{T}$, $\mu_r = 2$) was used to calibrate the Akoun-Yonnet analytical model, which yielded an effective magnetisation of $M_{\text{code}} = 2.20\,\text{MA/m}$, by matching the 40 T/m, 100 T/m and 300 T/m threshold-crossing distances to the COMSOL export data. Figure 7 shows that the calibrated analytical gradient profile agrees with COMSOL data to within 1% for the 40 T/m and 100 T/m threshold crossings, with the model agreement at 300 T/m being $\approx 12\%$, reflecting issues at the near-surface region of the strut: the Akoun-Yonnet model assumptions underestimate field concentration at strut corners and the FEM mesh

is within a coarser region at that scale. The root mean squared (RMS) error is 8.6 µm, which is below the 11 µm resolution of the 7-step binary search implemented in the trajectory analysis (Section 3.7) and therefore does not introduce any resolvable error in the capture distance predictions.



(a) (b)

Figure 7: Analytical model validation against COMSOL at $B_0 = 1.5\,\mathrm{T}$, $\mu_r = 2$. (a) Gradient profile $|\nabla|\mathbf{B}||$ as a function of arc length from the strut surface for all data models, the analytical curve uses $M = 2.20\,\mathrm{MA/m}$. (b) Threshold-crossing distances at 40 T/m, 100 T/m and 300 T/m, maximum discrepancy of 14.7 µm at 300 T/m ($< 12\%$).

The two models are independent, sharing no code and using different numerical kernels, whilst converging to the same gradient profile within an acceptable range over physiologically relevant distances. This agreement provides strong evidence that the gradient field used in the analytical framework is quantitatively reliable. The calibration was also verified to be independent of the SPION susceptibility model, since χ_{eff} (whether constant or using the Langevin saturation) only scales the force on the particle and does not affect the field computation.

4.2 Analytical Model Results

4.2.1 External Field and the Force Parameter

The analytical framework is used to examine the effect of the external uniform field $\mathbf{B}_0$ on the capture force, after calibration of the Akoun-Yonnet model against COMSOL (Section 4.1.4). The appropriate capture metric is the force parameter $F_p = |\mathbf{B}_{\text{tot}}| \cdot \nabla|\mathbf{B}_{\text{tot}}|$, which, as stated in Section 3.4.2, is proportional to the capture force on a SPION-labelled cell through Equation (11). The suppression of the scalar gradient and the subsequent enhancement of the force parameter is discussed in Section 3.3.2. Figure 8 analyses this enhancement of F_p, plotting it along the through-strut radial axis at $z = 0$.

Figure 8 analyses the force parameter, with panels (a) and (b) plotting F_p against distance from the strut surface over a range of $B_0 = \{0,\ 0.1,\ 0.5,\ 1.0,\ 1.5\,\mathrm{T}\}$ in the axial and transverse directions respectively. Despite the gradient suppression mentioned in Section 3.3.2, F_p is enhanced across all distances. For $B_0 = \{0,\ 0.1,\ 0.5,\ 1.0\,\mathrm{T}\}$ the values are approximately equal within the

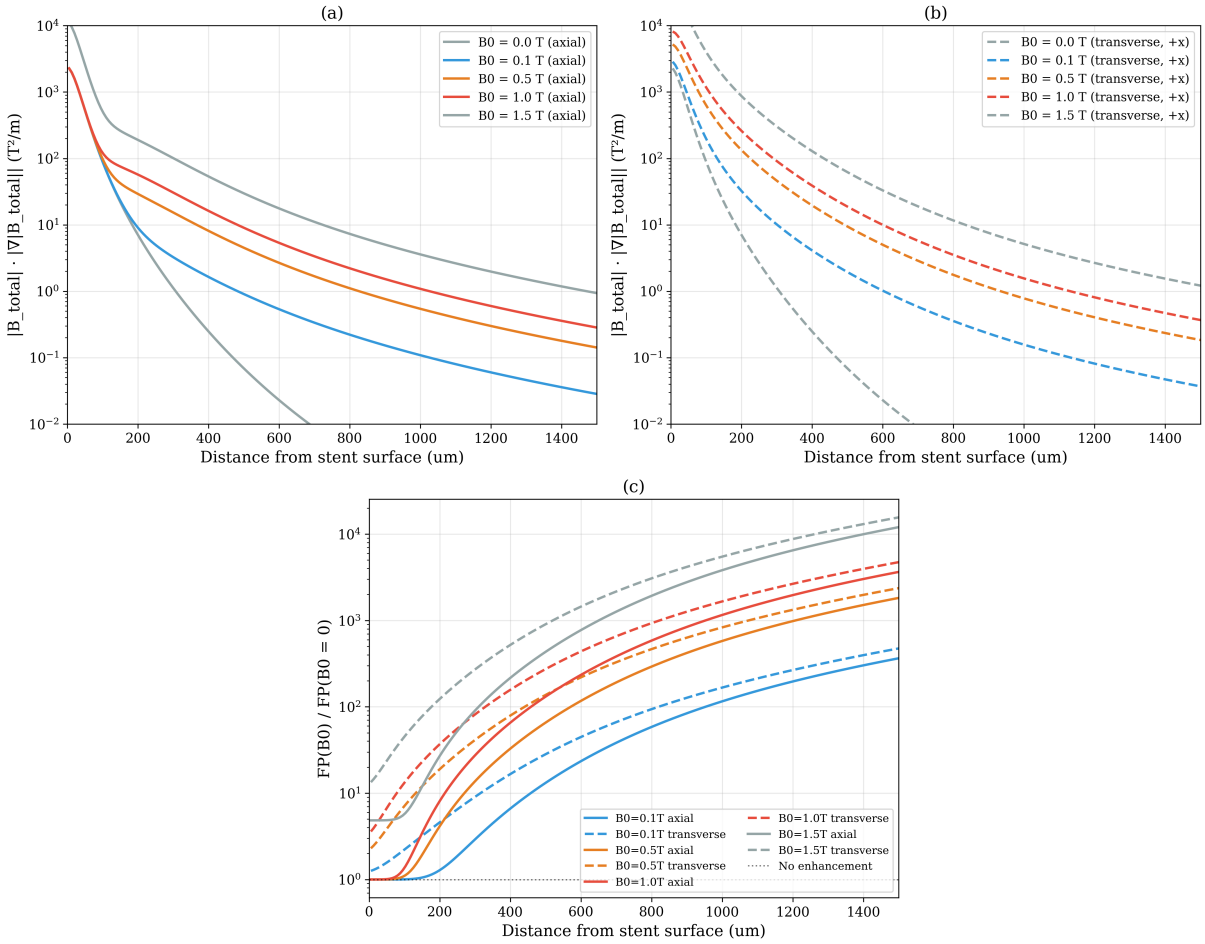


Figure 8: Force parameter $F_p = |\mathbf{B}_{\text{tot}}| \cdot \nabla|\mathbf{B}_{\text{tot}}|$ plotted as a function of distance from the stent surface in the axial (a) and transverse (b) directions. (c) Force parameter enhancement ratio.

first 100 µm in the axial direction, due to the stent's own field dominating over B_0.

Panel (c) quantifies the enhancement ratio $F_p(B_0)/F_p(B_0 = 0)$ against distance from the stent surface. The ratio increases with distance from the strut, and is minimal within the near-field ($d < 100$ µm), because at near-field distances the stent's own gradient dominates. In the far-field, the external field dominates and the enhancement grows to two to three orders of magnitude at $B_0 = 1.5$ T axial. The transverse configuration produces a stronger gradient and F_p than the axial configuration, however, clinical practice constrains the MRI field to the axial direction. The conclusion from this analysis is that F_p, and subsequently the capture range prediction, is an increasing function of B_0 regardless of the decrease in $|\nabla|\mathbf{B}||$. The full spatial decay of F_p is shown in Appendix A, Fig. A.3.

4.2.2 Static Capture Criterion and SPION Loading Sensitivity

The static capture criterion is built on the Furlani and Ng relation [53], $|F_{\text{mag}}| > |F_{\text{drag}}|$, and is applied across a wide SPION loading range (1–1000 pg Fe_3O_4) at four flow velocities: 0.05 m/s, 0.2 m/s, 0.4 m/s and 0.6 m/s, representing a range spanning diastolic, haemodynamically compromised flow to healthy MCA mean flow [51]. Figure 9(a) evaluates the static capture distance, defined as the maximum inward radial sweep distance satisfying the capture criterion, as a function of loading. Figure 9(b) plots the force ratio evaluated at 5 µm from the stent inner surface.

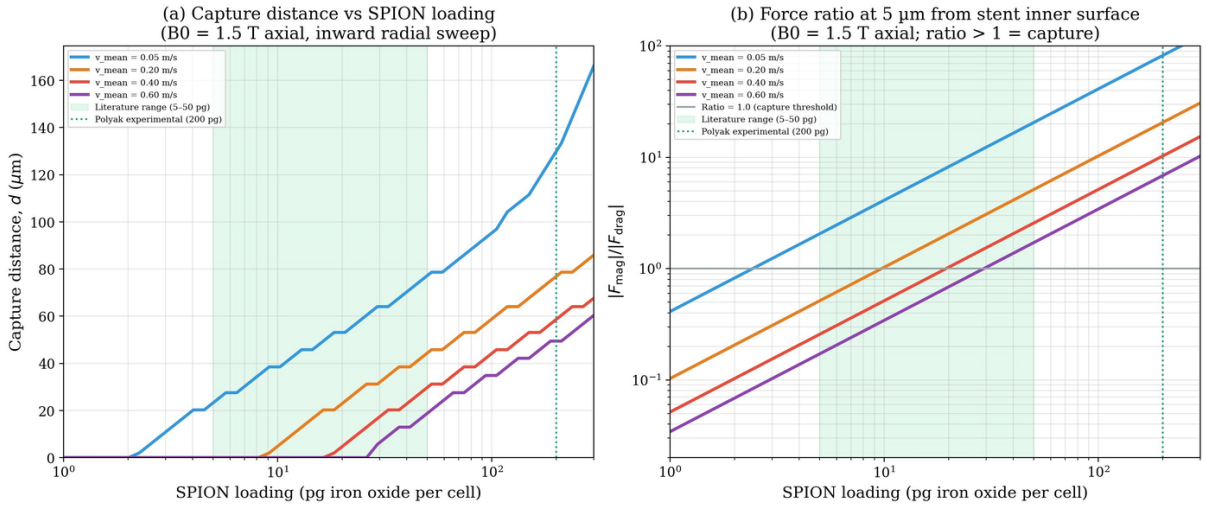


Figure 9: Static Furlani and Ng [53] capture criterion as a function of SPION loading (1–1000 pg) across four velocities (0.05 m/s, 0.2 m/s, 0.4 m/s, 0.6 m/s). (a) Maximum capture distance from stent surface. (b) Force ratio at 5 µm from stent inner surface.

A few observations can be made from these graphs. The static capture distance does not increase linearly with loading but instead follows an approximately logarithmic trend, representing the steep spatial decay of the gradient field, roughly decreasing by a factor of two per 50 µm. At higher velocities the rate of increase is more pronounced and exhibits a loading-threshold behaviour, reflecting the non-linear relationship between gradient field decay and Poiseuille flow. The capture distances across a loading range of 5–300 pg are presented in Table 5.

The data in Table 5 shows that capture is confined to a very narrow range in the near-wall region.

Table 5: Capture distance from stent inner surface (µm) as a function of SPION loading and mean blood velocity. $B_0 = 1.5\,\mathrm{T}$ axial, 12-strut stent, static Furlani and Ng criterion.

Loading (pg)	$\bar{v} = \mathbf{0.05\,m/s}$	$\bar{v} = \mathbf{0.20\,m/s}$	$\bar{v} = \mathbf{0.40\,m/s}$	$\bar{v} = \mathbf{0.60\,m/s}$
5	23.9	0.0	0.0	0.0
50	75.0	42.2	27.6	20.3
100	96.9	60.4	42.2	34.9
200	129.8	75.0	60.4	49.5

Across the typical literature loading range of 5–50 pg [61, 62] the capture range implies that a significant fraction of labelled cells would have zero static capture at relevant velocities. Even at 200 pg, the experimental optimal loading reported by Polyak et al. [8], the capture distance at healthy MCA flow $\bar{v} = 0.6\,\mathrm{m/s}$ [51] is 49.5 µm, whereas Polyak et al. [8] reported a capture efficiency of $\approx 20\%$ of circulating cells, implying a much greater effective capture distance.

4.2.3 Single Cell Trajectory and Drift Accumulation

The trajectory analysis integrates Equation (15) for a SPION-labelled cell injected 2 mm upstream of the stent ring model. Figure 10 implements the trajectory framework for two cells with loadings of 200 pg [8] and 10 pg under identical flow and field conditions, $\bar{v} = 0.05\,\mathrm{m/s}$ and $B_0 = 1.5\,\mathrm{T}$. Both cells are injected 160 µm away from the strut surface ($x = 1.30\,\mathrm{mm}$, $y = 0$, $z = -2\,\mathrm{mm}$). The 200 pg cell curves radially inwards during its transit and is captured, whilst the 10 pg cell drifts axially with negligible radial displacement and exits downstream.

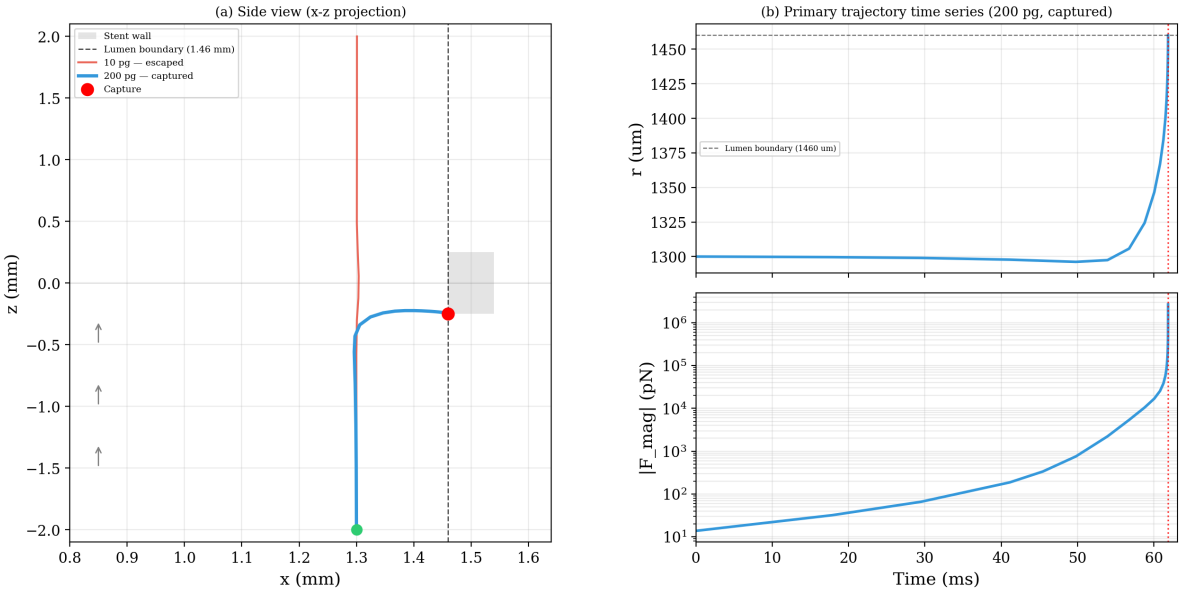


Figure 10: Single cell trajectory comparison between cells of loading 200 pg and 10 pg at $\bar{v} = 0.05\,\mathrm{m/s}$, $B_0 = 1.5\,\mathrm{T}$, injected 160 µm from the stent surface. (a) Side view projection (xz-plane). (b) Time series for the 200 pg cell trajectory: radial position $r(t)$ and magnetic force magnitude $|F_{\mathrm{mag}}|(t)$.

Figure 10(b) plots the trajectory of the 200 pg cell as a time series of radial position and force

magnitude $|F_{\text{mag}}|$. From this time series, $|F_{\text{mag}}|$ increases exponentially, rising from $\approx 50\,\text{pN}$ at injection to $\approx 10^3\,\text{pN}$ within $\approx 50\,\text{ms}$, corresponding to $z \approx -0.5\,\text{mm}$ from the stent strut. However, as seen in the time series of radial displacement, this force is insufficient to overcome the local Stokes drag and cause capture. Between $\approx 50\,\text{ms}$ and $\approx 62\,\text{ms}$ there is rapid radial acceleration as $|F_{\text{mag}}|$ rises by three orders of magnitude to $\approx 10^6\,\text{pN}$, overcoming the local Stokes drag and resulting in capture.

These results argue that capture cannot be defined through an instantaneous event at a single point using the static capture criterion $|F_{\text{mag}}| > |F_{\text{drag}}|$, but instead relies on modelling the cumulative effect of $|F_{\text{mag}}|$ on a cell through trajectory integration across an approach length. This is the origin of the trajectory versus static capture disparity discussed in Section 4.2.4 and represents one of the most important findings of the analysis.

4.2.4 Static Versus Trajectory Capture

The direct comparison of the static Furlani and Ng [53] capture prediction with the full trajectory integration over the physiologically relevant parameter space is presented in Figure 11 and Table 6. The plotted values represent the outermost initial radial injection point from which a strut-aligned cell is captured, resolved to 11 µm by a 7-step binary search. Two parametric sweeps are carried out: Figure 11(a) varies the SPION loading between 10–300 pg at $\bar{v} = 0.2\,\text{m/s}$, chosen to represent post-stenotic, diastolic flow and make the static vs. trajectory contrast more legible than at $\bar{v} = 0.6\,\text{m/s}$. Figure 11(b) varies the mean blood velocity from 0.02 m/s to 0.6 m/s at the Polyak et al. [8] experimental loading of 200 pg. Both figures use $B_0 = 1.5\,\text{T}$.

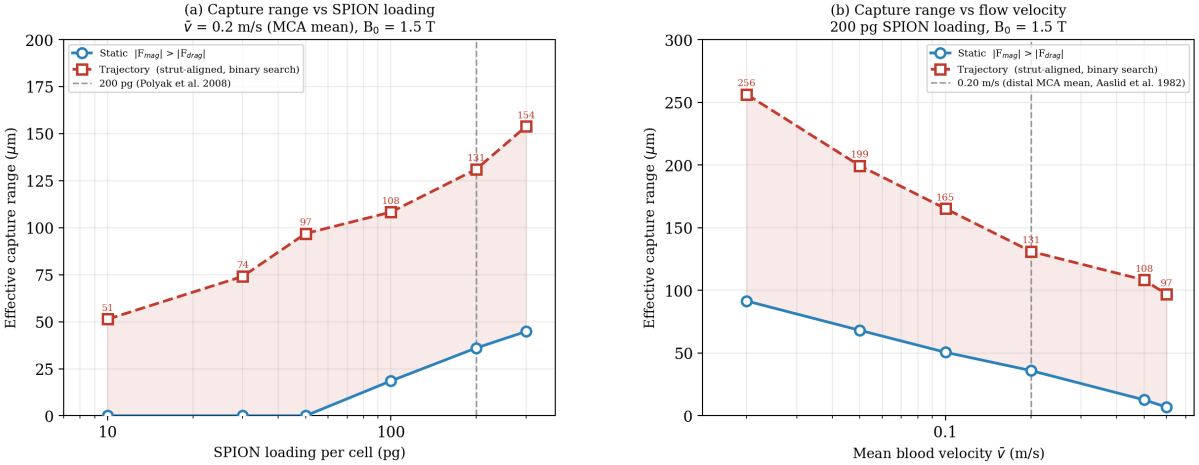


Figure 11: Static versus trajectory-integrated effective capture ranges at $B_0 = 1.5\,\text{T}$. Solid blue: static Furlani and Ng [53] criterion. Dashed red: trajectory-integrated criterion, 7-step binary search — 11 µm resolution. (a) Capture range versus SPION loading (10–300 pg) at $\bar{v} = 0.2\,\text{m/s}$. (b) Capture range versus mean blood velocity at 200 pg loading.

Figure 11 shows that the trajectory model predicts an effective capture range of 131 µm at $\bar{v} = 0.2\,\text{m/s}$, a 3.6× increase compared to the static criterion's 36 µm range. The disparity between the two models increases as loading decreases: at 50 pg the static criterion predicts zero capture whilst the trajectory integration has a 97 µm range (creating an infinite extension

factor). This reflects the loading dependence of the drift mechanism described in Section 4.2.3.

Figure 11(b) shows that the extension factor grows with flow velocity at the fixed 200 pg loading. At $\bar{v} = 0.02\,\mathrm{m/s}$, representing slow distal flow, the trajectory model predicts a capture range of 256 µm whereas the static model predicts 91 µm, giving an extension factor of 2.8×. The extension factors at $\bar{v} = \{0.05, 0.10, 0.20, 0.50, 0.60\,\mathrm{m/s}\}$ are presented in Table 6 and increase to 13.9× at 0.6 m/s. The local Stokes drag force scales linearly with velocity while the magnetic force is independent, meaning that the static criterion fails more severely at faster flows when compared to the trajectory model, which accumulates radial drift during the approach and whose radial impulse is less sensitive to velocity than the static criterion.

Table 6: Parameter sweep showing static capture distance, trajectory-integrated capture distance, and extension factor under varying loading and velocity conditions.

Parameter sweep	**Value**	**Static (µm)**	**Trajectory (µm)**	**Extension**
Loading ($\bar{v} = 0.2\,\mathrm{m/s}$)	10 pg	0	51	∞
Loading ($\bar{v} = 0.2\,\mathrm{m/s}$)	50 pg	0	97	∞
Loading ($\bar{v} = 0.2\,\mathrm{m/s}$)	100 pg	18	108	6.0×
Loading ($\bar{v} = 0.2\,\mathrm{m/s}$)	**200 pg**	**36**	**131**	**3.6×**
Loading ($\bar{v} = 0.2\,\mathrm{m/s}$)	300 pg	45	154	3.4×
Velocity (200 pg)	0.02 m/s	91	256	2.8×
Velocity (200 pg)	0.05 m/s	68	199	2.9×
Velocity (200 pg)	0.10 m/s	51	165	3.2×
Velocity (200 pg)	0.20 m/s	36	131	3.6×
Velocity (200 pg)	0.50 m/s	13	108	8.3×
Velocity (200 pg)	0.60 m/s	7	97	13.9×

The results of this analysis have two main implications. Firstly, the feasibility case for a cerebral magnetic retention device requires trajectory integration. Secondly, the capture efficiency of ≈ 20% reported by Polyak et al. [8] is consistent with the drift-accumulation model demonstrated here, supporting the physical reliability of the trajectory framework as a preliminary analysis method for ascertaining capture ranges.

4.2.5 Population Capture Efficiency and Literature Benchmarking

The effective capture range presented in Section 4.2.4 does not quantify the efficiency of cell capture from an injected cell population. Figure 12 extends the trajectory integration analysis to a population of 20 cells injected at $z = -2\,\mathrm{mm}$ along a radial, strut-aligned line spanning the magnetically active near-wall field, $r = 1.20\,\mathrm{mm}$ to $1.45\,\mathrm{mm}$. Each trajectory ODE (15) is integrated independently. The population capture efficiency η_{cap} is defined as the fraction of cells captured across the integration window.

Figure 12(a) displays η_{cap} as a function of mean blood velocity at the Polyak et al. 200 pg loading [8]. Efficiency is 100% at $\bar{v} \leq 0.02\,\mathrm{m/s}$, where the capture zone spans the full 250 µm injection band, and decreases monotonically to 40% at $\bar{v} = 0.5\,\mathrm{m/s}$. The reduction in capture efficiency is non-linear in velocity: doubling the velocity from 0.1 m/s to 0.2 m/s reduces η_{cap}

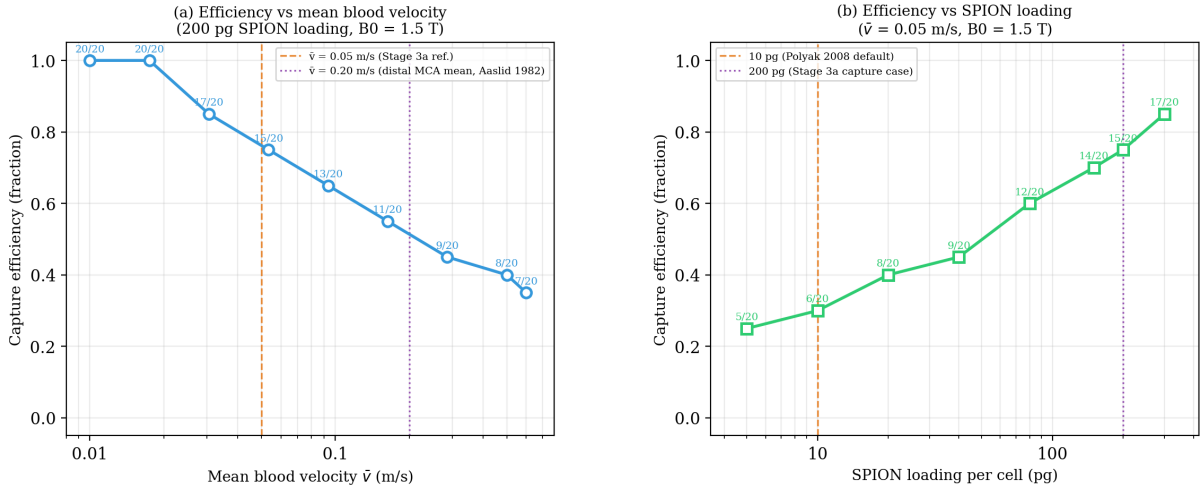


Figure 12: Population capture efficiency from 20 cells injected along a radial, strut-aligned azimuth, $B_0 = 1.5\,\mathrm{T}$. (a) Capture efficiency against mean blood velocity at 200 pg SPION loading. (b) Capture efficiency against SPION loading.

from 65% to 55% — a 1.2× reduction compared to the 2× reduction expected from Stokes drag scaling — which represents the population-level signature of the radial drift accumulation mechanism.

Figure 12(b) displays η_{cap} as a function of SPION loading at $\bar{v} = 0.05\,\mathrm{m/s}$, chosen to create a clear interpretable comparison. This curve rises monotonically with an approximately logarithmic shape from an efficiency of 25% at 5 pg to 85% at 300 pg. This further enforces that increased SPION loading directly increases the capture zone radially, with diminishing effect as the capture front advances into the weaker far-field region. The increase in capture efficiency with SPION loading is supported by in-vivo experiments where magnetised stents provided significantly more capture of SPION-loaded cells [8, 41]. It should be noted that the model uses a small cell population to reduce computation time, and that the population is strut-aligned along the $\theta = 0$ azimuth where gradient is strongest, which does not account for the random distribution of a real cell population. The gradient heatmaps in Section 4.1.2 show a significant decrease in gradient field strength between struts, meaning that angular-averaged efficiencies over the strut period will be lower. The angular-averaged sweep is identified as future work.

5 Discussion, Limitations and Future Work

5.1 Interpretation of the Static-versus-Trajectory Capture Gap

The most significant finding during this project's duration was the underestimation of effective capture range predicted by the static Furlani and Ng force-balance criterion [59] relative to the implemented full trajectory integration. Using the experimental loading from Polyak et al. [8] at representative, post stenotic conditions (200 pg, $\bar{v} = 0.2\,\mathrm{m/s}$, $B_0 = 1.5\,\mathrm{T}$) the static criterion predicted a capture range of 36 µm, compared to the 131 µm effective capture range predicted by the trajectory-integrated model — a 3.6× underestimate. This failure is not a minor numerical

discrepancy but rather a qualitative error that would lead to the conclusion that the proposed ferromagnetic stent device cannot function effectively at relevant, clinical SPION loadings.

The single-cell trajectory analysis, section 4.2.3, identifies this mechanism. In the model, a cell injected 160 µm from the stent strut surface experiences a magnetic force in the order of 50 pN at injection which is insufficient to overcome the local Stokes drag. However, over the 2 mm upstream approach, the cell accumulates radial inward drift as it traverses regions of increasing magnetic field gradient, placing the cell within the capture boundary layer within the final ≈0.5 mm of approach as $|F_{\text{mag}}|$ rises by ≈three orders of magnitude. This cannot be achieved by the static criterion as it retains no memory of the cells upstream history and drift.

The implementation of trajectory based analysis has been implemented in studies of stent-assisted magnetic targeting [42, 63] with interparticle interactions using sub-micron carrier particles. The model used throughout this project analyses the effectiveness of this model at MCA relative conditions with MSC geometry parameters to assess the feasibility of a neuro-therapeutic device.

5.2 Capture Range, SPION Loading and Clinical Relevance

The capture range defined by the 200 pg loading spans 131 µm radially into the vessel, representing a capture zone of ≈ 9% of the lumen radius. Although this represents a small, near-wall region, the SPION loaded cells can take advantage of margination if the MSCs are relatively small, or if micro drug carriers are used to deliver therapeutic effects through paracrine signalling [64].

The logarithmic relationship between capture range and SPION loading presented in Figure 11 reflects the steep decay in the gradient of the magnetic field. This creates practical design consequences, where the most efficient loading range is between 50–200 pg, the range at which incremental loading produces the greatest marginal increase in effective capture range. This is also useful as increasing SPION loadings are potentially cytotoxic [65] and would impair the cell or drug carriers therapeutic effect for a minimal increase in capture range.

The cell population capture efficiency of ≈ 55% at 200 pg, $\bar{v} = 0.2$ m/s, represents an upper bound on the spatially averaged efficiency, as all cells were aligned with the strut, experiencing the maximum gradient field. The gradient heatmaps produced in COMSOL reveal that the field is weaker between adjacent struts, meaning that realistic capture efficiencies would be lower as cells approach from azimuths between the struts would experience smaller magnetic forces. If there was a spatial average over the full strut period, this could potentially reduce the reported efficiencies by 50% or more, bringing the spatially averaged efficiency closer to the 20% in-vivo efficiency reported by Polyak et al. [8], however this cannot be claimed and verified without additional analysis.

5.3 Gradient Decay, Stent Geometry and the Null-Gradient Zone

The COMSOL analysis demonstrated that the radial component of the magnetic gradient decay from a stent strut follows a power law $G(d) = A \cdot d^n$ with an exponent $n \approx -1.25$ to -1.34, shallower than the far-field dipole decay of r^{-3} predicted analytically. This behaviour is expected as the relevant observation range, 25 µm to 300 µm, is comparable in distance to the strut dimensions themselves ($w = 100\,\mu m$, $t = 80\,\mu m$), meaning that the far-field regime is not satisfied. This fitted power law is more relevant for the actual gradient decay over the physiologically relevant range and the capture zone prediction model than utilising the theoretical far-field expression.

Although the exported data required relatively significant cleansing, the consistency of the exponent n across the four stent models to within $\approx 7.7\%$ confirms that the strut count predominantly governs the azimuthal coverage of the capture zone. This creates another design implication where increasing strut count or coverage does not meaningfully increase the radial capture distance, but does increase the circumferential fraction of the lumen wall that falls within the effective capture zone.

The null-gradient zone identified in V2 and V3 heatmaps (5 and 6) arises due to the intrinsic consequence of $\nabla|\mathbf{B}|$ being a function of constituent field vectors. When the gradient fields from two adjacent struts overlap, the tangential components partially cancel, collapsing the scalar gradient magnitude and creating an effective 'dead zone' for SPION capture. This phenomenon also creates a design implication, where an optimal design would balance the azimuthal coverage against these null-gradient regions that emerge between struts when the inter-strut spacing becomes comparable to the gradient decay length.

5.4 Saturation Model and Robustness

The implementation of the Langevin saturation correction reduces the effective SPION susceptibility by approximately 5.7× at $B_0 = 1.5\,T$ in comparison to the linear regime ($\chi_{\text{eff}} \approx 0.35$ vs $\chi_0 = 2.0$). This is a significant correction that would heavily overestimate the magnetic force on a cell if ignored. A cross-check with constant χ shows that this only propagates to a $\approx 20\%$ increase in the trajectory-integrated capture range as the model is already in a near-saturated regime and the capture predictions are dominated by stent geometry and the spatial structure of the gradient field.

Limitations

Steady Poiseuille Flow

The most significant limitation in the hydrodynamic model is the simplification of cerebral, pulsatile flow to steady Poiseuille flow. In reality, cerebral arteries are subject to pulsatile flow with a Womersley number of $\alpha \approx 2$ at average heart rate of $1Hz$ (60 bpm). This means that the quasi-steady approximation relatively reasonable as the instantaneous velocity profile

does not deviate substantially and time-averaging errors are estimated at 10–20%. However, this pulsatility does introduce a time-varying drag force that could either transiently assist or oppose radial drift accumulation depending on when the cell arrives at the stent during the cardiac cycle.

The implementation of a pulsatile flow model would replace the mean-velocity parameter with a time-resolved velocity field and produce a more physically accurate representation of capture probability and is the most important hydrodynamic refinement for the model. Furthermore, the Stokes drag is also an approximation when considering the cells full trajectory and the use of Oseen drag and Faxén corrections would yield higher fidelity [66, 67].

Single-Ring Analytical Model

The analytical model reduces the three-dimensional braided stent geometry to a single planar ring of n rectangular struts at $z = 0$. This simplification isolates the dominant capture physics to the axial cut-plane, where peak gradient occurs, which is justified by the COMSOL analysis. Although the axially displaced strut segments contribute significantly less to the local gradient as observed in the COMSOL model, the multi-ring periodicity of the braided stent geometry is not represented. The periodicity of the the stent means that, in reality, cells approaching along the vessel axis pass through multiple successive gradient field zones. The superposition of these contributions would extend the effective approach length and therefore increase accumulated radial drift. This means that the single-ring model used in the analytical framework produces a conservative lower bound on the capture range compared to a full three-dimensional geometry. The use of an arbitrary $\mu_r = 2$ value could also be improved by implementing the full $\mathbf{B} - \mathbf{H}$ curve of a selected ferromagnetic material.

No Cell–Cell Interactions or Brownian Motion

All of the cell trajectories were integrated independently, assuming that cell populations do not alter the flow field or the magnetic force experienced by neighbouring cells. For a clinically injected dose, the cell population is a sufficiently low fraction of the total cell volume in the lumen meaning that this assumption is appropriate for preliminary analysis. However, the accumulation of cells on the stent surface could shadow the gradient field experienced by other cells approaching the stent, which is not represented in this model.

Linear Magnetic Permeability of the Stent

The COMSOL model uses an assigned linear relative permeability value of $\mu_r = 2$ for the stent material, representing the saturated state of a soft ferromagnetic steel. The model does not incorporate the full $\mathbf{B}-\mathbf{H}$ curve, which represents the relationship between magnetic flux density $\mathbf{B}$ and magnetic field strength $\mathbf{M}$. This means that the spatial variation of magnetisation within the struts is not captured self-consistently. This assumption propagates to the analytical model, where the effective magnetisation $M_{\text{eff}} = 2.20\,\text{MA/m}$ was calibrated against the COMSOL

gradient profile. The analytical model therefore overestimates the stent contribution at lower external fields, $\mathbf{B}_0$ and underestimate at higher fields where the stent is more deeply saturated.

No Shear-Induced Detachment Model

The analysis only addresses the initial capture event but does not model the retention of captured cells under continuous shear. Captured cells adhering to the stent surface experience a hydrodynamic lift force and wall shear stress that act to detach the cell. The wall shear stress at baseline conditions is $\tau_w = 4\eta\bar{v}/R_{\text{ves}} \approx 2.1\,\text{Pa}$, which sits in the atheroprotective range (¿1.5Pa) [68] but could be sufficient to displace weakly adhered cells. A complete analysis of therapeutic retention would require a model of the cell-surface adhesion probability under physiological shear.

Angular-Averaged Capture Efficiency

The most useful extension to the current capture efficiency model would involve the computation of the angular-averaged population efficiency by modelling multiple cell injection points across the full 2π azimuth. This implementation would replace the current through-strut upper bound with a more clinically relevant spatial mean and better quantify the realistic capture efficiencies, which would be expected to be lower than the current reported values and is essential for credible therapeutic dose prediction.

6 Conclusions

This dissertation project set out to analyse the feasibility of ferromagnetic stent retention device to capture engineered, SPION labelled, cells for targeted therapeutic delivery in cerebral arteries. The main focus was to define a capture range and analyse how the magnetic gradient field decays across the vessel lumen. The initial stages involved extensive literature review of the restrictive properties of the BBB and relevant transport mechanisms, which presented the question of whether magnetism could enable effective cell therapy delivery to the brain. The analysis proceeded through COMSOL finite-element modelling of four braided stent geometries, an analytical framework built upon the Akoun-Yonnet expression for the magnetic flux density of a uniformly magnetised rectangular prism which was cross-validated against the COMSOL results, and a trajectory ODE integration model for characterising the effective capture range and population efficiency.

The principal conclusion is that trajectory integration is essential for a minimum level of physical fidelity required to predict capture performance at cerebral flow conditions. The primary mechanism responsible for cell capture is the cumulative radial drift of the cells across the approach distance, which brings the injected cells within the steep near-strut gradient region and explains the discrepancy between the static capture criterion and the trajectory-integrated model.

The COMSOL analysis established that the gradient field decay follows a fitted power law with

exponent $n \approx -1.25$ to -1.34 and is insensitive to strut count across the models. This verifies that the gradient field responsible for radial capture distance is determined primarily by the strut geometry and magnetisation rather than by the total number of struts. The strut count does however govern the azimuthal coverage of the capture zone, with null-gradient zones being identified between adjacent struts in higher-porosity models being an intrinsic limitation of the stent architecture due to $\nabla|\mathbf{B}|$ being a function of constituent field vectors.

The trajectory-integrated capture range spans 36 µm to 256 µm across the velocity range, $\bar{v} = 0.02, 0.05, 0.1, 0.2, 0.5, 0.6 m/s$ at a 200 pg SPION loading and $B_0 = 1.5\,\mathrm{T}$, with population efficiency falling monotonically from 100% at $\bar{v} = 0.02\,\mathrm{m/s}$ to 35% at $\bar{v} = 0.6\,\mathrm{m/s}$. Establishing that (even when accounting for an assumed 50% decrease in efficiency when regarding the implementation of angular-averaged population efficiency – which would bring the reported capture efficiency to $\approx 25\%$ at MCA conditions) cerebral magnetic stent retention of SPION-loaded cells is feasible under physiological flow.

The analysis demonstrates that a strong external magnetic field is necessary, with the default 1.5 T MRI-strength background field providing meaningful capture ranges. The external field saturates the SPION loading which eliminates the susceptibility uncertainty, drives the stent into the saturation magnetisation regime and enhances the force parameter $|\mathbf{B}_{\mathrm{tot}}| \cdot \nabla|\mathbf{B}_{\mathrm{tot}}|$ by two to three orders of magnitude. The MRI environment is therefore an integral part of the capture physics, not just a convenient clinical context.

The main limitations of the analysis include: steady Poiseuille flow, single-ring axial geometry, strut-aligned angular sampling, and the absence of a shear-induced detachment model. The single-ring axial geometry is a conservative assumption, underestimating the capture range relative to a full three-dimensional braided geometry, whereas the strut-aligned angular sampling assumption overestimates the capture range. The steady Poiseuille flow assumption is more ambiguous as a pulsatile flow model could either transiently assist or oppose radial drift accumulation and therefore increase or decrease the effective capture range depending on the cells location relative to the cardiac cycle.

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Appendix A Supplementary Figures

A.1 Finite-difference gradient convergence

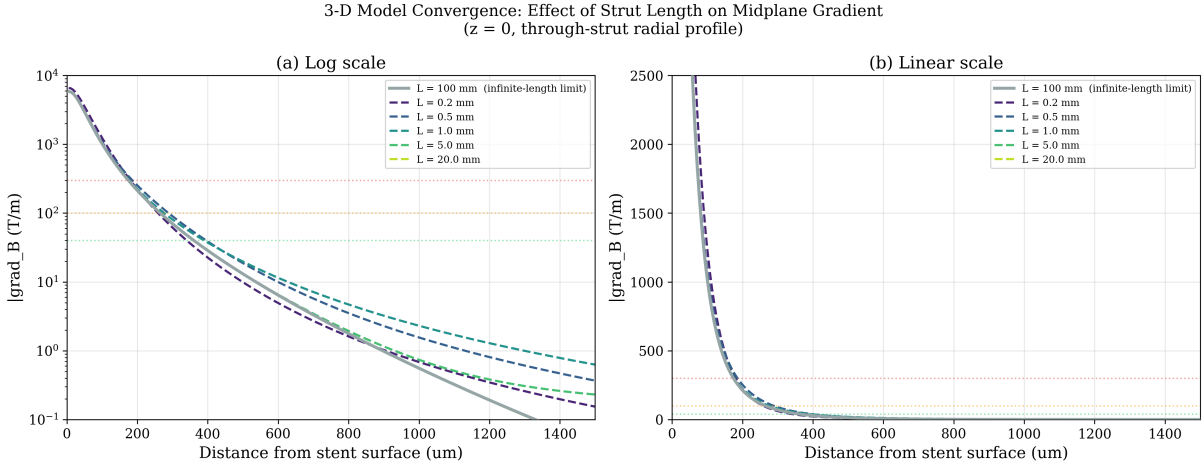


Figure 13: Convergence of the finite-difference gradient computation. Halving the grid spacing Δx reduces the gradient-magnitude error quadratically, confirming $\mathcal{O}(\Delta x^2)$ scaling.

A.2 Hydrodynamic drag characterisation

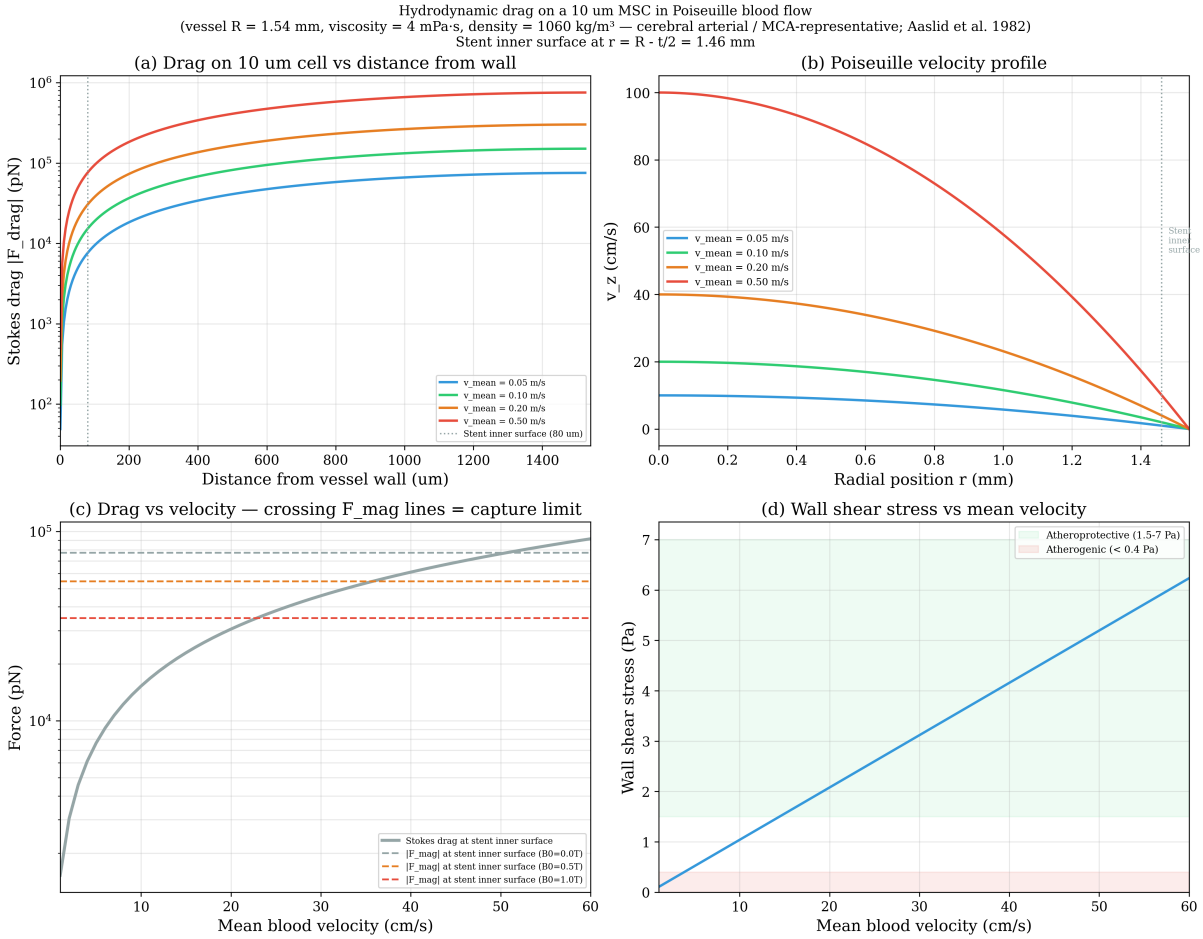


Figure 14: Hydrodynamic drag on a 10 µm diameter cell under Poiseuille flow. **(a)** Stokes drag versus radial position for four mean flow velocities (0.02–0.5 m s^{-1}). **(b)** Poiseuille velocity profile at $\bar{v} = 0.2$ m s^{-1}. **(c)** Drag versus mean velocity, with magnetic-force thresholds. **(d)** Wall shear stress versus mean velocity.

A.3 Force parameter spatial decay

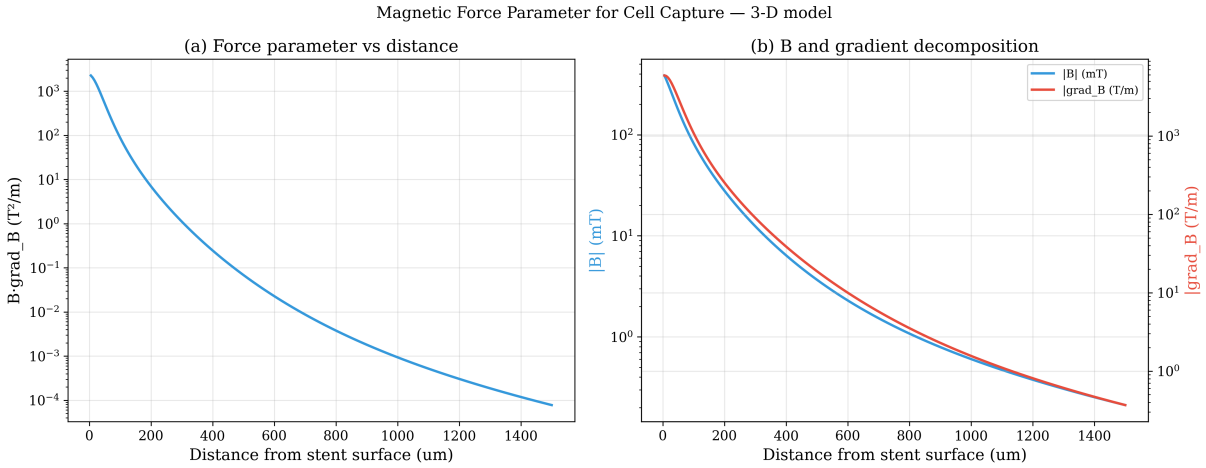


Figure 15: Force parameter $|\mathbf{B}| \cdot |\nabla|\mathbf{B}||$ as a function of radial distance from the stent surface along the through-strut axis.The individual $|\mathbf{B}|$ and $|\nabla|\mathbf{B}||$ contributions is shown in (b)

Appendix B Software Architecture

The analytical framework is implemented in Python and is called `Project_Code`. The code is used to generate all the necessary figures through the methods described in this report.

B.1 Directory layout

```
Project_Code/
|-- requirements.txt
|-- run_all_figures.py              # regenerates figures into results/
|-- data/
|   |-- comsol_gradient_data_cleaned.xlsx  # V1-V4 datasets
|   `-- V1-Mgrad-Plot-New2.csv             #V1  updated export
|-- results/                        # created on first run
`-- stent_capture/                  # Python package
    |-- __init__.py
    |-- core/
    |   |-- field_model.py          # StentRing (Akoun & Yonnet)
    |   `-- gradient.py             # finite-difference |grad|B||
    |-- physics/
    |   |-- external_field.py       # UniformExternalField + TotalField
    |   |-- magnetic_force.py       # SPIONLabelledCell, Langevin
    |   |-- hydrodynamics.py        #Poiseuille Flow + Stokes drag
    |   `-- capture_criterion.py    # static |F_mag| > |F_drag|
    |-- simulation/
    |   |-- trajectories.py         # integrate_trajectory (RK45)
    |   `-- capture_efficiency.py   # injection-line & parameter sweeps
    |-- data/
    |   `-- comsol_loader.py        # openpyxl-based .xlsx reader
    `-- figures/
        |-- common.py               # DEFAULTS, calibration, make_ring
           ()
        |-- style.py                # matplotlib rcParams
        |-- fig07_comsol_gradient_validation.py
        |-- fig07extra_comsol_multigeometry.py
        |-- fig08_force_parameter.py
        |-- fig09_spion_loading_sweep.py
        |-- fig10_single_trajectory.py
        |-- fig11_static_vs_trajectory.py
        `-- fig12_capture_efficiency.py
```